AgriCycle: A Community-Based Biomass Information Management and Market Linkage System

A Thesis

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CHAPTER 1
INTRODUCTION

1.1 Background of the Study

Agricultural biomass refers to crop and processing residues such as rice straw, rice husks, coconut husks, coconut shells, corn stalks, corn cobs, palm oil residues, banana pseudostems, sugarcane bagasse, wood residues, and other farm by-products. These materials may support composting, biochar, animal feed where suitable, industrial heating, grain drying, domestic cooking, renewable energy, and other waste-to-value activities. In many local settings, however, their value is not lost simply because the residues have no possible use. It is lost because information about quantity, location, timing, present use, condition, and availability is incomplete, scattered, or not validated in a shared information system.

This study is focused on Butuan City because the city has already identified biomass as one possible renewable energy resource under the Butuan City Energy Development Plan 2023-2050. The BEDP presents Butuan as a growing agro-industrial and smart city that needs cleaner and more secure energy options. It also sets a direction toward net-zero emissions by 2050 through renewable energy, energy efficiency, electric mobility, and stronger local energy planning (City Government of Butuan, 2023).

The first biomass figures for Butuan appear strong. Based on 2019 agricultural data, the BEDP identified rice, coconut, corn, and palm oil residues as possible feedstock sources. The total theoretical feedstock was estimated at 216,536.2 metric tons per year, with a theoretical energy potential of 3,278 terajoules or 910,665 megawatt-hours per year. Using 25 percent plant efficiency and 80 percent capacity factor, the estimate was translated into a possible capacity of about 32.49 megawatts (City Government of Butuan, 2023; Preferred Energy, Inc., 2025).

The important issue is that theoretical biomass is not the same as usable biomass. A residue may be counted in a city-level estimate, but it may already be used by a rice mill, scattered across distant farms, too wet for storage, available only for a short period, or too costly to collect. Therefore, this study does not treat biomass as available only because it appears in an estimate. For research purposes, practical

biomass availability must be understood as the result of several information conditions: present use, source location, timing, storage, quality observation, aggregation readiness, and stakeholder verification.

The biomass pre-feasibility study for Butuan shows this gap clearly. Rice straw is found in fields after harvest and is often burned when farmers prepare for the next planting. In Butuan, rice harvesting is still mainly manual, making rice straw collection tedious and labor-intensive. Rice husks are easier to locate because they are found in rice mills, but Butuan City staff reported that rice husks are already used for industrial heating, grain drying during the rainy season, and domestic cooking. Coconut husks may be used as fuel for copra kilns, while coconut shells may be processed into charcoal. Corn and palm oil residues account for a smaller share of theoretical potential and are scattered in different locations (Preferred Energy, Inc., 2025).

Wood residues provide the clearest quantitative example of the difference between total supply and practical availability. The Q4 2024 woodwaste survey recorded 1,063.62 metric tons of wood residues per month, but 991.25 metric tons were already used by the companies themselves. Only 72.37 metric tons were not used. This means more than 93 percent of surveyed wood residues were already internally used, mostly for process heating. The pre-feasibility study therefore concluded that Butuan did not have enough collectable feedstock to sustainably support even a 1-megawatt biomass power plant unless a dedicated feedstock source is developed (Preferred Energy, Inc., 2025).

This finding does not weaken the need for AgriCycle. It makes the information systems research problem clearer. The issue is not only whether biomass exists, but how stakeholders can identify which biomass remains practically available after existing uses and field conditions are considered. Without this information management layer, stakeholders may overestimate supply, plan around unverified feedstock, duplicate records, delay decisions, or miss smaller waste-to-value opportunities that could still be coordinated locally.

Table 1. Biomass Potential and Practical Availability Constraints in Butuan City

Biomass Source	Known Quantitative Basis	Practical Availability Constraint
Rice residues	About 106,072 MT/year of feedstock, 2,011 TJ/year, and 558,570 MWh/year theoretical potential. The pre feasibility study also identified 37 rice mills. AGRAC was reported at 500 MT/day rice production, equal to about 140 MT/day rice husk using a 0.28 husk ratio.	Rice straw is field-based, seasonal, scattered, and costly to collect under manual harvesting. Rice husks are concentrated in mills but may already be used for heating, grain drying, and domestic cooking.
Coconut residues	About 39,725 MT/year of feedstock, with 945 TJ/year and 262,634 MWh/year theoretical potential	Husks and shells are generated at farms and during copra processing, but husks may already be used for copra kilns, while shells may be processed into charcoal. Actual availability depends on present use, storage, location, and collection timing.
Corn residues	About 34,074 MT/year of feedstock, with 310 TJ/year and 86,133 MWh/year theoretical potential.	Corn residues have a smaller share compared with rice and coconut and are found in different locations. Aggregation may be difficult without barangay-level or cooperative-level records.
Palm oil residues	About 166 MT/year of feedstock, with 12 TJ/year and 3,328 MWh/year theoretical potential.	The quantity is small compared with rice and coconut. These residues may still be useful for specific users, but they should not be overstated in city-level biomass planning.
Wood residues	The Q4 2024 survey recorded 1,063.62 MT/month total wood residues: 991.25 MT/month used by companies and 72.37 MT/month not used.	More than 93 percent of surveyed wood residues were already internally used. This shows that total residue production cannot be treated automatically as available supply.

Table 1 shows that Butuan City has biomass potential, but each biomass source has a practical availability constraint. For AgriCycle, this means that biomass records should not stop at total estimated production. The system must help separate total production, present use, reserved quantity, and available quantity. It must also support records on source location, availability period, storage condition, quality observation, and validation status before any residue is treated as available for aggregation or buyer inquiry.

The economic value of biomass also depends on information. Agricultural residues can become inputs for heat, energy, compost, biochar, and other biomass-based products, but market value is difficult to reach when supply is not

organized and verified. IRENA (2023) noted that agricultural residues are often limited by missing supply chains, storage concerns, preprocessing needs, and the need for year-round supply. Nunes and Silva (2023) also emphasized that logistics costs affect whether residual biomass can be used economically. In Butuan, these concerns are visible because residues exist, but they are scattered, seasonal, partly used, and not yet organized into a reliable information flow.

1.1.1 Stakeholder Ecosystem of Biomass Availability Management

The biomass problem in Butuan is also a stakeholder ecosystem problem. Different actors hold different parts of the information. Farmers and processors know when residues are produced. Rice mills know how much husk is generated and used internally. Cooperatives may know where supply can be consolidated. Buyers know what biomass they need. LGU offices need records for planning, reporting, and waste-to-value decisions. The problem begins when these pieces remain separated.

Table 2. Stakeholder Ecosystem and Information Flow Needed by AgriCycle

Stakeholder	Current Information Role	Observed Information Gap	Needed Link in AgriCycle
Smallholder farmers and biomass contributors	Generate residues from farming activities and may know when residues are available after harvest.	Records may be informal, unstandardized, or not encoded in a shared system. Some farmers may also have limited direct access to digital platforms.	Report biomass through cooperative hubs, rice mills, or authorized users who can encode and validate records.
Rice millers and agricultural processors	Generate or handle residues such as rice husks, coconut residues, and processing by products.	Present use may not be visible to buyers or LGU offices. This can create false assumptions about available supply.	Record total production, present use, reserved quantity, available quantity, and storage or quality observations.
Cooperative hubs and aggregation centers	Can consolidate information and physical supply from smaller sources.	Manual records may not clearly show aggregation status, storage condition, or updated availability.	Serve as local data entry, validation, inventory, and aggregation points.
Institutional buyers and biomass users	Need reliable information on quantity, location, quality, and availability period before making inquiries.	Supply uncertainty may discourage inquiry because listings may be unverified, seasonal, or already used.	View validated listings and submit inquiries based on recorded location, quantity, status, and quality notes.

LGU offices and planning stakeholders	Need updated records for agricultural mapping, BEDP monitoring, waste-to value planning, and future resource assessment.	Data may be fragmented across offices, surveys, and stakeholders, making planning less timely and less precise.	Use reports, dashboards, GIS views, and audit logs to support monitoring and local planning.

Table 2 clarifies why the study cannot be treated only as a software development project. The research issue is the missing coordination layer among the people who produce, consolidate, use, and plan around biomass. The desired flow is not farmer to buyer directly in all cases. A more realistic local flow is farmer or processor to cooperative hub, then to AgriCycle, then to a validated inquiry-based market linkage and LGU monitoring. This makes the system community-based and keeps the scope still aligned with biomass availability information management.

1.1.2 Why Butuan City is the Study Site

Butuan City was selected as the study site because it has an existing local energy planning direction, documented biomass potential, and pre-feasibility findings showing the difference between theoretical feedstock and practical collectable supply. These local conditions make Butuan a suitable setting for studying biomass availability information management. The detailed justification for selecting Butuan City, including the structured LGU comparison and related selection criteria, is discussed in Chapter 2. In this chapter, Butuan is introduced only as the local context where AgriCycle Butuan will be designed, developed, and evaluated.

1.1.3 Research Gap Analysis

The research gap of this study is not only a feature gap. It is a knowledge and information management gap. Existing biomass studies often estimate potential, analyze spatial distribution, study logistics, or propose market linkage. These are important, but they provide limited mechanisms for verifying practical availability at the community level. In Butuan, the question is not simply how much biomass may exist. The stronger research question is how practical biomass availability can be identified, validated, mapped, aggregated, and shared among local stakeholders through an information system.

Table 3. Research Gap Analysis for AgriCycle Butuan

Literature or Source Group	Main Focus	Knowledge Gap for this Study	Response of AgriCycle
Spatial biomass resource assessment studies (Bharti et al., 2021; Galang et al., 2021; Ukoba et al., 2024)	Estimate biomass potential, map spatial distribution, or identify suitable biomass energy sites.	They show where biomass may be located, but they provide limited explanation of how community-level actors can verify whether mapped biomass is still available after present use, storage, and timing are considered.	Uses GIS as part of practical availability management, not only potential mapping. Source locations are linked with records on present use, availability status, storage, and hub monitoring.
Biomass supply chain and logistics studies (Chidozie et al., 2023; IRENA, 2023; Nunes & Silva, 2023)	Discuss supply chain costs, collection, storage, transport, preprocessing, and resource availability concerns.	They explain why biomass is hard to coordinate, but they do not translate these constraints into a local information model for cooperative data entry, validation, mapping, and buyer inquiry.	Converts supply chain constraints into system fields and workflows: source location, availability period, storage condition, quality notes, aggregation status, and validation status.
Biomass market linkage studies (Borisoot et al., 2024)	Connect biomass suppliers and users through online matching or market linkage functions.	They support buyer supplier matching, but they do not fully address present-use verification, cooperative-hub validation, and LGU reporting before biomass is shown to buyers.	Limits the market linkage module to validated listings and inquiries, reducing the risk of false, duplicate, or unavailable supply.
Digital agriculture systems (Trendov et al., 2019; AgriBOOST, 2022)	Support digital transformation, agricultural information sharing, market access, and rural service delivery.	They are broad and often focused on crop production, farm services, inputs, or produce marketing, not residue availability and waste-to-value coordination.	Applies digital agriculture to a focused local problem: biomass availability information management through authorized users and hubs.
Butuan biomass and local energy planning documents (City Government of Butuan, 2023, 2024; Preferred Energy, Inc., 2025)	Identify theoretical biomass potential and practical feedstock limitations in Butuan City.	They provide strong local evidence, but they are not a continuing information system that updates records after the study period.	Develops a working system that can record, update, validate, map, and share practical biomass availability information.

Therefore, the study generates knowledge by identifying the critical factors that affect community-level biomass availability management and translating them into the Community Based Biomass Availability Information Management Model (CB-BAIM Model). The expected contribution is not only the system. It is also a

focused model showing how biomass information should move from source, to hub, to validation, to buyer inquiry, and to LGU planning support.

1.1.4 Contribution to Circular Economy

AgriCycle contributes to circular economy thinking by improving the information stage of material recovery. Circular economy does not only mean reusing waste. It also means making materials visible, traceable, and usable within a system that can support recovery and recirculation. Agricultural waste valorization supports circular economy goals because residues can become useful inputs instead of being burned, left unmanaged, or treated as useless materials (Ellen MacArthur Foundation, n.d.; Ufitikirezi et al., 2024).

In this study, the circular economy contribution is focused and realistic. AgriCycle does not claim to directly produce compost, biochar, energy, income, or emission reduction. Its role is earlier in the chain: it organizes the records needed before residues can be recovered and reused. By connecting production data, current use, condition, aggregation, and inquiry, the system supports resource efficiency, waste reduction, material recirculation, and more careful biomass utilization through information management.

Table 4. Circular Economy Pathway Supported by AgriCycle

Circular Economy Step	Meaning in this Study	AgriCycle Support
Residue generation	Biomass is produced from farming, milling, copra processing, or other agricultural activities.	Records biomass type, source, estimated quantity, and availability period.
Recovery identification	Stakeholders identify which residues are unused, underused, reserved, or already used.	Separates total production, present use, reserved quantity, and available quantity.
Aggregation	Small and scattered residues are consolidated through hubs, mills, or authorized users.	Supports cooperative-hub records, inventory updates, storage status, and validation.
Inquiry-based market linkage	Validated biomass may be shown to possible users for inquiry.	Provides validated listings, buyer inquiry records, and inquiry status monitoring.

Reuse or waste-to-value use	Biomass may be directed to compost, biochar, heat, energy, research, or other legitimate uses where suitable.	Provides reports and availability data for planning, while leaving actual conversion and environmental impact for future studies.

This pathway makes the circular economy role of the system clearer. AgriCycle supports material recirculation by improving the data that allows residues to be identified, checked, aggregated, and offered for possible reuse. It does not replace physical collection, processing, or feasibility studies. Instead, it provides the verified information needed before those activities can be planned properly.

1.2 Statement of the Problem

This study aims to design, develop, and evaluate AgriCycle: A Community-Based Biomass Information Management and Market Linkage System. It addresses the fragmented recording, validation, mapping, aggregation, and sharing of biomass availability information in Butuan City. It also supports the proposed Community-Based Biomass Availability Information Management Model (CB-BAIM Model).

Specifically, this study seeks to answer the following questions:

1. What core problems affect biomass availability management in Butuan City in terms of:
   1.1 Data Collection, including how biomass quantity, source, current use, timing, condition, and storage information are recorded;
   1.2 Information Management, including how records are standardized, validated, updated, mapped, and classified; and
   1.3 Stakeholder Coordination, including how farmers, processors, cooperative hubs, buyers, and LGU offices share and use biomass records for aggregation and waste-to-value planning?
2. How may the CB-BAIM Model be translated into AgriCycle as a web-based system through:
   2.1 Biomass data capture and validation;
   2.2 Cooperative-hub aggregation and inventory management;
   2.3 GIS-based source mapping and inquiry-based market linkage coordination; and

2.4 Reporting, inquiry status monitoring, and audit logging?

3. How does the developed AgriCycle system perform when evaluated by intended users in terms of:

3.1 Functional Suitability, including the completeness, correctness, and usefulness of the system functions; and

3.2 Usability, including ease of use, learnability, interface clarity, and efficiency of user interaction?

1.3 Objectives of the Study

The general objective of this study is to design, develop, and evaluate AgriCycle: A Community-Based Biomass Information Management and Market Linkage System. The study also formalizes the CB-BAIM Model as a practical guide for local biomass data flow, validation, aggregation, buyer inquiry, and planning support.

Specifically, this study aims to:

1. To examine the core problems affecting biomass availability management in Butuan City in terms of:

1.1 Data Collection, focusing on biomass quantity, source, current use, timing, condition, and storage information;

1.2 Information Management, focusing on record standardization, validation, updating, mapping, and classification; and

1.3 Stakeholder Coordination, focusing on the information flow among farmers, processors, cooperative hubs, buyers, and LGU offices.

2. To design and develop AgriCycle based on the CB-BAIM Model through:

2.1 Biomass data capture and validation;

2.2 Cooperative-hub aggregation and inventory management;

2.3 GIS-based source mapping and inquiry-based market linkage coordination; and

2.4 Reporting, inquiry status monitoring, and audit logging.

3. To evaluate the developed AgriCycle system using selected ISO/IEC 25010 software quality characteristics in terms of:

3.1 Functional Suitability, including the completeness, correctness, and usefulness of the system functions; and

3.2 Usability, including ease of use, learnability, interface clarity, and efficiency of user interaction among intended users.

1.4 Significance of the Study

This study is significant because biomass planning in Butuan City needs updated and practical data. The BEDP provides a renewable energy direction, while the pre-feasibility study shows that actual collectable feedstock is more limited than theoretical estimates. AgriCycle responds as an information system that focuses on what biomass exists, what is already used, what remains available, where it is located, when it can be collected, and who may inquire about it.

Smallholder farmers and biomass contributors may benefit because their residues can become visible through cooperative hubs or authorized users. They do not need to manage online accounts directly, which is important in areas where digital access and digital skills may vary (Trendov et al., 2019).

Rice millers and agricultural processors may benefit because the system separates total residue production from internal use and remaining availability. This reduces the risk of false listings and supports more accurate coordination with buyers and LGU offices.

Cooperative hub managers may benefit because the system gives them a clearer role in data entry, validation, inventory monitoring, aggregation status updates, and buyer coordination. This supports the community-based structure of the study.

Institutional buyers and biomass users may benefit because the system provides validated listings with quantity, location, availability period, present use status, storage condition, and quality observations. This can reduce uncertainty before buyers make inquiries.

The Local Government Unit of Butuan City may benefit because the system can support agricultural mapping, BEDP monitoring, waste-to-value planning, coordination with cooperatives, and future biomass assessments. Shared and updated information may help local offices avoid repeated data requests and improve planning based on practical availability rather than theoretical estimates alone.

Environmental and sustainability stakeholders may benefit because the study supports the information stage of resource recovery and waste reduction. This is aligned with the intent of Republic Act No. 9003 on ecological solid waste management, Republic Act No. 9513 on renewable energy, Republic Act No. 11285 on energy efficiency and conservation, and selected Sustainable Development Goals, particularly SDG 7, SDG 9, SDG 11, SDG 12, and SDG 13 (United Nations, 2015).

Future researchers and system developers may use the study as a basis for further work on biomass availability factors, cooperative-based data management, GIS-based biomass monitoring, inquiry-based market linkage systems, and waste-to-value information models.

1.5 Scope and Delimitations of the Study

Scope of the Study

This study focuses on the design, development, and evaluation of AgriCycle: A Community-Based Biomass Information Management and Market Linkage System. The system supports organized recording, validation, mapping, inventory monitoring, buyer inquiry coordination, reporting, and information exchange for biomass resources in Butuan City.

The study covers agricultural biomass and related residues within the Butuan City setting, including rice straw, rice husks, coconut husks, coconut shells, corn residues, banana pseudostems, sugarcane bagasse, palm oil residues, wood residues where relevant, and similar materials. It also considers existing uses such as industrial heating, grain drying, domestic cooking, copra kiln fuel, charcoal production, process heating, composting, biochar, animal feed use where applicable, and other legitimate waste-to-value applications.

The system includes modules for biomass data collection and management, cooperative based aggregation, GIS-based biomass mapping, inquiry-based market linkage, inquiry status monitoring, audit logging, and reports or dashboards. The primary users are cooperative hub managers, rice mill personnel, aggregation center personnel, authorized LGU representatives, cooperative coordinators, institutional buyers, biomass information users, and system administrators. Smallholder farmers

are treated as offline biomass contributors whose information may be encoded by authorized users.

The developed system will be evaluated using selected characteristics of the ISO/IEC 25010 Software Quality Model, specifically Functional Suitability and Usability. Functional Suitability will assess whether the system provides the needed functions. Usability will assess whether the system is easy to learn and use for intended users (ISO/IEC, 2011).

Limitations of the Study

This study is limited to the development and evaluation of a web-based information system. It does not establish a city-wide biomass collection program, physical aggregation facility, storage warehouse, biomass processing plant, or power plant.

The study does not claim that Butuan City currently has enough collectable biomass to support a biomass power plant. The pre-feasibility study is used as a basis for strengthening the information needed of the system, not as proof of immediate power project feasibility.

The study does not perform economic feasibility analysis, power plant design, route optimization, transport cost optimization, fleet management, real-time vehicle tracking, or logistics simulation. Transport-related data are included only as records that may affect coordination.

The study does not perform laboratory testing or scientific certification of biomass quality. Quality information is limited to manually recorded field observations such as moisture condition, cleanliness, storage condition, visible contamination, and other practical concerns.

The inquiry-based market linkage component supports listings, buyer inquiries, contact coordination, and inquiry status tracking only. It does not include digital payment processing, online banking, automated procurement, bidding, contract management, price negotiation automation, legally binding transactions, or transaction settlement.

The GIS component is limited to mapping and location-based visualization. It does not include real-time GPS tracking, satellite monitoring, or advanced spatial prediction.

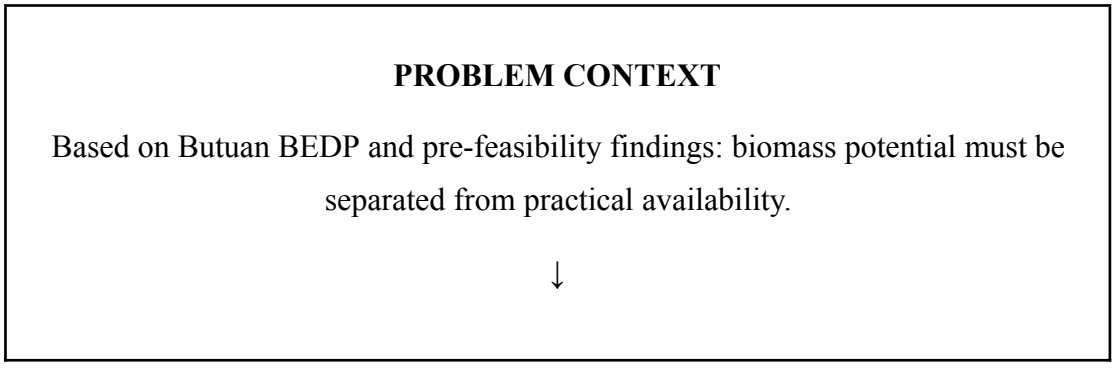
The evaluation is limited to available respondents, experts, and stakeholders during the study period. It focuses only on Functional Suitability and Usability. Other software quality characteristics such as security, reliability, maintainability, compatibility, portability, and performance efficiency are outside the evaluation scope.

The study does not measure long-term economic impact, actual farmer income increase, actual emission reduction, biomass utilization efficiency, renewable energy output, or policy effectiveness. These may be studied after the system is deployed and after enough verified records are collected.

1.6 Research Framework

The research framework shows how the biomass information problem in Butuan City is translated into the CB-BAIM Model and the design and evaluation of AgriCycle. It is grounded on the Butuan City Energy Development Plan 2023-2050 and the biomass pre feasibility study, which show that biomass potential must be checked against practical collectable supply (City Government of Butuan, 2023; Preferred Energy, Inc., 2025). It also uses selected theories and models to connect information quality, stakeholder workflow, supply coordination, GIS mapping, circular economy direction, and software quality evaluation (Bostrom & Heinen, 1977; DeLone & McLean, 2003; Mentzer et al., 2001; Densham, 1991; ISO/IEC, 2011).

Figure 1. Research Framework of the Study



INTEGRATED THEORETICAL AND CONCEPTUAL ANCHORS

Socio-Technical Systems Theory | Information Systems Success Model | Biomass Supply Chain Concept | Spatial Decision Support / GIS | Circular Economy Theory

↓

SYSTEM DESIGN RESPONSE

Biomass data capture | Availability validation | Present-use documentation | Storage and quality notes | Cooperative-hub aggregation | GIS mapping and visualization | Buyer inquiry coordination | Reports and audit logs

↓

DEVELOPED SYSTEM OUTPUT

AgriCycle: A Community-based biomass availability information management system

↓

EVALUATION LENS

ISO/IEC 25010: Functional Suitability and Usability

↓

EXPECTED CONTRIBUTION

Biomass availability factors and information management model for aggregation, buyer inquiry, and LGU planning

The framework begins with the local problem context because AgriCycle is not designed from a general biomass issue alone. The BEDP establishes that Butuan has biomass potential, while the pre-feasibility study shows that this potential must be checked against present use, location, timing, storage, quality, and transport concerns (City Government of Butuan, 2023; Preferred Energy, Inc., 2025). This explains why the system focuses on practical biomass availability instead of simple biomass estimation.

The theories and concepts in the framework work together rather than separately. Socio Technical Systems Theory explains the people-process-technology link. The Information Systems Success Model explains why the records must become useful, accurate, and accessible information. The Biomass Supply Chain Concept explains why records must reflect handling, storage, quality, aggregation, and movement from source to hub or buyer. Spatial Decision Support explains why location and distance affect the usefulness of biomass records. Circular Economy Theory explains why residues are treated as recoverable resources instead of unmanaged waste. Together, these anchors form the logic of the CB-BAIM Model.

These anchors lead directly to the system design response. Data capture responds to fragmented records. Availability validation responds to the difference between estimated and usable supply. Cooperative-hub aggregation responds to scattered sources and varied digital access among farmers. GIS mapping responds to location-sensitive supply. Buyer inquiry coordination and reporting respond to buyer uncertainty and LGU planning needs. This flow links the local problem, the academic basis, and the system functions in one direction.

The evaluation lens is placed after the developed system output because ISO/IEC 25010 is not a theory and not a system feature. It is used to determine whether the developed system is functionally suitable and usable. Functional Suitability checks whether the system provides the needed functions for recording, validation, mapping, aggregation, inquiry, reporting, inquiry status monitoring, and audit logging. Usability checks whether intended users can learn and use the system clearly and efficiently (ISO/IEC, 2011).

Overall, the framework shows that AgriCycle is a research-centered information system study. It does not only produce a system. It proposes the CB-BAIM Model by identifying the factors and workflows needed to manage biomass information from source reporting to hub validation, buyer inquiry coordination, and LGU planning support.

1.7 Novelty of the Study

The novelty of AgriCycle lies in its focus on community-based biomass availability management, not only biomass potential, sustainability claims, or system

features. Existing biomass studies commonly identify source locations, estimate possible supply, study storage and transport issues, or connect suppliers and buyers through digital platforms. These are useful, but they do not fully answer the local research problem in Butuan: how biomass information can be verified, organized, mapped, aggregated, and shared at the community level.

This study treats biomass availability as an information management problem. A biomass record becomes meaningful only when it can support decisions on source, quantity, current use, condition, collection timing, aggregation, inquiry, and planning. This shifts the study from a simple system feature gap to a knowledge contribution: the CB-BAIM Model.

Another novel aspect is the community-based cooperative-hub model. Instead of requiring every farmer to directly use the system, authorized users such as cooperative hub managers, rice mill personnel, LGU representatives, NGO coordinators, or institutional users may encode, validate, and update biomass records. This makes the system more suitable for local conditions where access to devices, internet connection, and digital skills may vary (Trendov et al., 2019).

The market linkage feature is also availability-based and inquiry-based. AgriCycle does not simply allow biomass to be posted for buyers. It links inquiry with supporting records on source, quantity, current use, condition, location, and availability period. This reduces the risk of showing false, duplicate, or unavailable supply while keeping the system outside the scope of payment, selling, or legally binding transactions.

Therefore, the novelty of AgriCycle is its integrated and local approach. It combines availability validation, GIS-based mapping, cooperative-hub aggregation, buyer inquiry coordination, and LGU reporting into one Butuan-specific information management system. Its contribution is the CB-BAIM Model, a practical information model that turns scattered biomass records into organized, checked, mapped, and usable data for agricultural waste-to-value management.

Table 5. Novelty Matrix of AgriCycle Based on Reviewed Studies

Literature Basis with Authors / Sources	Main Focus and Relevance	Gap Addressed by AgriCycle
Spatial Biomass Resource Assessment Studies - Bharti et al. (2021); Galang et al. (2021); Ukoba et al. (2024).	Focus on locating biomass sources, mapping resource distribution, and estimating biomass potential using GIS or spatial analysis. These studies support the GIS mapping component because biomass sources in Butuan are spread across farms, rice mills, barangays, and possible collection points.	These studies show where biomass may be located, but they do not fully verify whether the biomass is still available, already used, properly stored, or ready for aggregation and buyer inquiry.
Biomass Feedstock Supply and Handling Studies - Chidozie et al. (2023); IRENA (2023); Nunes and Silva (2023).	Focus on biomass collection, storage, transport, handling, quality, seasonality, and supply chain concerns. These studies support the need to record storage conditions, quality notes, aggregation status, collection timing, and practical availability.	These studies explain supply and handling concerns, but they do not provide a local information system that can record, validate, map, and coordinate biomass availability among cooperatives, buyers, and LGU offices
Biomass Resource Matching and Market Linkage Studies - Borisoot et al. (2024)	Focus on connecting biomass suppliers with possible buyers, users, or bioenergy facilities through digital platforms or market linkage models. This supports the buyer inquiry feature because available biomass should be visible to possible users.	Market linkage studies support buyer-supplier matching, but they do not fully address local validation, present-use documentation, cooperative-hub data entry, LGU reporting, and Butuan-specific biomass conditions.
Digital Agriculture Information System Studies - Trendov, Varas, and Zeng (2019); AgriBOOST (2022)	Focus on the use of digital tools and platforms in agriculture, including information sharing, rural services, market access, and digital adoption issues. This supports the digital and community based structure, especially the use of authorized users or hubs.	These studies usually focus on crop production, farm services, inputs, or produce marketing. They do not directly focus on biomass residues, availability validation, aggregation monitoring, and waste-to-value coordination.
Butuan Biomass and Local Energy Planning Documents - City Government of Butuan (2023, 2024); Preferred Energy, Inc. (2025)	Focus on Butuan City's renewable energy direction, biomass potential, and practical feedstock limitations. These sources justify why the system is focused on Butuan City.	These documents provide planning direction and biomass evidence, but they do not provide a working system for updated records, validation, GIS mapping, cooperative aggregation, buyer inquiry coordination, and LGU reporting.

Table 5 shows that AgriCycle is not novel because of one feature alone. Its novelty comes from combining several functions that are usually separated in existing studies and linking them to one local information management problem. GIS-based studies support mapping, but do not fully verify actual availability. Supply and handling studies explain storage, quality, and transport concerns, but do not usually

provide a local information workflow for stakeholder coordination. Market linkage studies support buyer-supplier matching, but do not fully check whether the listed biomass is already used, unavailable, or unsuitable. Digital agriculture studies support technology use in farming, but are usually focused on crops, inputs, services, or produce marketing.

AgriCycle addresses these limits by treating biomass as a record that must be checked before it becomes a listing, report, or planning input. This is the study's main knowledge contribution: it defines biomass availability as a managed information condition, not merely a physical residue estimate. The system is the application of that contribution through the CB-BAIM Model.

1.8 Definition of Terms

Actual biomass availability. biomass that is not only produced but can still be collected, recorded, stored, or offered after considering present use, location, timing, and condition.

Agricultural biomass. plant-based residues from farming or processing activities, such as rice straw, rice husks, coconut husks, coconut shells, corn stalks, banana pseudostems, and similar materials.

Availability status. the classification of a biomass record as available, already used, reserved, unsuitable, pending validation, or uncertain.

Biomass aggregation. the consolidation of biomass records or physical supply from different farms, rice mills, barangays, collection points, or cooperative hubs.

Biomass availability information. organized data showing biomass type, quantity, location, availability period, present use, storage status, quality notes, validation status, and buyer inquiry status.

Biomass quality information. field-level observations about the condition of biomass, such as moisture, cleanliness, storage condition, visible contamination, or other practical concerns.

Biomass Availability Information Management Model. the research output that organizes the factors and workflow needed to record, validate, map, aggregate, and share practical biomass availability information.

Cooperative-hub model. the system setup where authorized cooperative managers, rice mill personnel, LGU representatives, NGO coordinators, or other institutional users handle encoding and validation while farmers may participate offline.

Functional Suitability. the degree to which the system provides the needed functions, including completeness, correctness, and usefulness of functions.

GIS mapping. the mapping feature used to show biomass source locations, rice mills, cooperative hubs, collection points, and other relevant places in Butuan City.

Inquiry-based market linkage module. the system module where validated biomass listings may be viewed by possible buyers and where inquiries may be submitted for coordination. It does not include payment, buying, selling, bidding, contract signing, or transaction settlement.

Market linkage. the information-based connection between biomass providers, cooperative hubs, and possible buyers. In this study, it is limited to validated listings, inquiries, and coordination.

Present use. the current use of biomass before it is listed as available, such as rice husks used by rice millers for drying or heating.

Practical availability. the portion of biomass that may realistically be used after considering existing use, seasonality, collection timing, distance, storage, and quality condition.

Research gap. the part of the problem that previous studies or systems have not fully addressed, which this study aims to respond to.

Usability. the degree to which the system is easy to learn and use, including interface clarity, learnability, and efficiency of interaction.

CHAPTER 2

REVIEW OF RELATED LITERATURE

This chapter reviews the theories, studies, and related systems that support a Community-Based Biomass Information Management and Market Linkage System. The chapter is written around the main contribution of the study, the Community-Based Biomass Availability Information Management (CB-BAIM) Model. The model treats biomass availability as an information system problem. It does not assume that agricultural residues are available only because they appear in production reports, resource estimates, or GIS maps.

The chapter uses one clear flow that is consistent with Chapter 1 and will guide Chapter 3: source identification, data capture, community-based validation, information management, and coordination use. This sequence is important because the study is not mainly about estimating biomass volume. It is about turning scattered residue information into checked, mapped, and usable decision information for cooperatives, buyers, and LGU offices.

To avoid a broad and unclear theoretical base, this review separates the foundations into three levels. Design Science Research is the main foundation because the study develops and evaluates a system artifact. Information Quality and Data Quality Theory is the core problem theory because the contribution depends on credible and usable biomass availability records. The other theories and concepts support the system design by explaining stakeholder roles, supply conditions, location, waste-to-value direction, and software evaluation.

2.1 Comparative Basis for Selecting Butuan City as the Study Scope

The selection of Butuan City as the study scope was based on a comparative LGU checklist. The checklist was based on existing legal, technical, environmental, agricultural, and digital governance bases. This was done to avoid choosing a place only because it is agricultural. Instead, the selected LGU must show actual biomass residue streams, clear coordination needs, possible value-chain users, environmental alignment, local governance support, and enough available evidence.

The comparison included Butuan City, Davao City, Cagayan de Oro City, South Cotabato Province, Bukidnon Province, and the Davao del Norte/Tagum area. These places were considered because they have public evidence related to agriculture, biomass, organic waste recovery, renewable energy, farm logistics, or market coordination. Some of the comparison areas are provinces rather than cities because biomass data and agricultural residue studies are often documented at the provincial level, especially when the concern involves crop production, residue distribution, and possible aggregation.

Table 6. Basis of the LGU Selection Criteria

Selection Criterion	Basis of the Criterion	Reason for Inclusion
Specific agricultural biomass residue streams are documented	Republic Act No. 9513 (2008) recognizes biomass resources as biodegradable organic materials, including agricultural by-products and residues.	The selected LGU must have actual residue streams, not only general agriculture activity.
Residues are tied to multiple biomass types, not only one crop	Villar et al. (2016) used agricultural reports and survey data to estimate biomass availability and energy potential from corn residue in Bukidnon.	A biomass information system becomes more useful when residue types can be classified by source, location, and possible use.
There is evidence of aggregation, collection, logistics, or coordination gaps	The City Government of Cagayan de Oro (2025) recognized the need to identify gaps in localized logistical support for farmers and improve coordination with key partners.	Biomass is bulky and location-based; therefore, collection and aggregation problems must be considered in selecting the study area.
LGU has formal agriculture programs or agriculture planning evidence	Republic Act No. 7160 (1991) devolves agricultural support services and local development functions to LGUs.	The selected LGU must have an agriculture office or related programs that can support data validation and possible adoption.
Farmer/community participation structure is visible	Republic Act No. 7160 (1991) recognizes the role of local government units and barangays in basic services and agricultural support.	A community-based study needs farmer groups, barangays, or local actors who can provide and verify information.
Environmental/resource recovery alignment is visible	Republic Act No. 9003 (2001) supports ecological solid waste management, composting, recycling, and materials recovery facilities.	Agricultural residues should be treated as possible recoverable resources, not only as waste.
Renewable-energy or biomass development alignment is visible	Republic Act No. 9513 (2008) promotes renewable energy development, including biomass. DA-CRAO/AMIA also promotes	The selected LGU should have a sustainability direction related to biomass, waste-to-value use, or climate-resilient agriculture.

	climate-resilient agriculture and local planning support.	
Digital agriculture or market coordination precedent exists	Trendov, Varas, and Zeng (2019) explain that digital technologies can support agriculture and rural value chains. DICT (2024) also promotes digital transformation of public services.	Digital readiness strengthens the possibility of using an information system for agriculture-related coordination.
Information-system intervention is directly justified	Peffers et al. (2007) explain that design science research begins with problem identification and proceeds to the design, development, demonstration, and evaluation of an artifact.	The selected LGU must show a real information or coordination problem that can be addressed through a system-based study.
Potential buyers or off-takers can be identified	Republic Act No. 9003 (2001) supports composting and recovery, while Republic Act No. 9513 (2008) recognizes agricultural residues as biomass resources.	Biomass has practical value only if there are possible users such as composters, processors, nurseries, farms, mills, or energy-related actors.

Table 6 shows that each criterion is supported by an outside basis such as law, technical biomass assessment, agriculture governance, environmental policy, digital agriculture, and research feasibility. This makes the checklist more defensible because it avoids choosing an LGU based only on preference or location.

The first group of criteria focuses on biomass resource evidence. Republic Act No. 9513 (2008) recognizes agricultural residues and by-products as biomass resources, which means that the selected LGU must show actual residue streams such as rice husks, rice straw, coconut husks, coconut shells, corn residues, or other farm-based materials. This is important because the study cannot be based only on general agriculture, it must be based on biomass residues that can be recorded, located, and assessed.

The next group of criteria focuses on aggregation, logistics, and spatial distribution. Biomass is usually bulky, seasonal, and scattered, so the presence of residue alone is not enough. The area must also show a need for collection points, clustering, mapping, or coordination. This is supported by technical biomass assessment practices, where quantity, location, current use, and available potential are considered before biomass can be treated as usable supply.

The environmental and sustainability criteria are also important. Republic Act No. 9003 (2001) supports composting, recycling, materials recovery, and waste diversion, while Republic Act No. 9513 (2008) supports biomass as part of renewable

energy development. These laws support the idea that agricultural residues should not be viewed only as waste. They may also become recoverable materials for composting, processing, energy-related use, or other waste-to-value activities, depending on their actual condition and availability.

The governance and community criteria strengthen the role of the LGU. Republic Act No. 7160 (1991) gives LGUs important functions in agriculture support, local development, and barangay-level services. This means the selected area must have offices, barangays, farmer groups, or local actors that can help provide and validate data. This is necessary because an information system will fail if no local office or community actor can update records, confirm biomass availability, or use the reports.

The digital governance and information-system criteria make the checklist more aligned with the study's field. Trendov, Varas, and Zeng (2019) explained that digital technologies can support agriculture and rural value chains, while Peffers et al. (2007) emphasized that design science research should begin with a clear problem and produce a solution that can be evaluated. Therefore, the chosen LGU must not only have biomass; it must also have a clear information or coordination gap that can reasonably be addressed through a system.

Overall, Table 6 gives the study a strong foundation because the selection criteria are based on recognized policies and research logic. It proves that the LGU screening is not random. It also shows that the study is not simply asking where biomass exists, but where biomass information, aggregation, market linkage, and LGU monitoring need to be improved.

Table 7. Comparative LGU Checklist Matrix

No.	Selection Criterion	Butuan City	Davao City	Cagayan de Oro City	South Cotabato Province	Bukidnon Province	Davao del Norte / Tagum Area
1	Specific agricultural biomass residue streams are documented	Yes	Partial	Partial	Partial	Yes	Yes
2	Residues are tied to multiple biomass types, not only one crop	Yes	Partial	Partial	Partial	Partial	Partial
3	There is evidence of aggregation, collection, logistics, or coordination gaps	Yes	Partial	Partial	Partial	Partial	Partial
4	LGU has formal agriculture programs or agriculture planning evidence	Yes	Yes	Yes	Yes	Partial	Partial
5	Farmer/community participation structure is visible	Yes	Yes	Partial	Partial	Partial	Partial
6	Environmental/resource recovery alignment is visible	Yes	Yes	Partial	Yes	Partial	Partial
7	Renewable-energy or biomass development alignment is visible	Yes	Partial	Partial	Yes	Yes	Yes
8	Digital agriculture or market coordination precedent exists	Yes	Partial	Partial	No	No	No
9	Information-system intervention is directly justified	Yes	Partial	Partial	Partial	Partial	Partial
10	Potential buyers/off-takers can be identified	Partial	Partial	Partial	Yes	Partial	Partial

Table 7 presents the comparative analysis of Butuan City, Davao City, Cagayan de Oro City, South Cotabato Province, Bukidnon Province, and the Davao del Norte/Tagum area. These areas were included because each has available public evidence related to agriculture, biomass resources, organic waste recovery, renewable-energy direction, farm logistics, or agricultural market coordination. Provinces such as South Cotabato, Bukidnon, and Davao del Norte are included because biomass evidence is often reported at the provincial level, especially when the concern involves crop production, residue distribution, and energy potential.

The table shows that Butuan City has the most complete alignment across the checklist. Preferred Energy, Inc. and WWF Philippines (2025) identified rice, coconut, corn, and palm oil as biomass sources in Butuan City, including rice straw and husks, coconut husks and shells, corn stalks, cobs and husks, and palm oil shells, fiber, and empty fruit bunches. The same report also estimated 216,536.2 MT/year of total biomass feedstock and 3,278 TJ/year of theoretical energy potential using 2019 data. These findings give Butuan a strong biomass basis because the residues are documented, varied, and supported by a formal pre-feasibility assessment.

Butuan City also shows a clear link between biomass potential and long-term local planning. The Butuan City Energy Development Plan 2023–2050 identifies biomass as part of the city's renewable-energy direction, while related city energy planning references connect Butuan's energy roadmap to a zero-net emission goal by 2050 (City Government of Butuan, 2023; Preferred Energy, Inc. & WWF Philippines, 2025). This strengthens Butuan's position because its biomass evidence is not isolated, it is connected to a broader local energy and sustainability direction.

Davao City was considered because it has an established local policy direction on organic agriculture and resource recovery. The Davao City Organic Agriculture Ordinance of 2012 supports the use of biodegradable waste as a resource for organic agriculture. This gives Davao City a strong governance and environmental basis. However, based on the checklist, its available public evidence is rated only partial in biomass residue documentation and renewable-energy or biomass development alignment.

Cagayan de Oro City was considered because it has agriculture planning evidence related to mechanization, farm-to-market roads, and logistical support. The

City Government of Cagayan de Oro (2025) identified the need to improve localized logistical support for farmers and strengthen coordination with key partners. This makes the city relevant for agriculture governance and logistics criteria. However, its available public evidence is still more focused on general agriculture modernization than on clearly documented agricultural biomass residue streams.

South Cotabato Province was considered because it has public evidence on organic waste-to-resource and waste-to-energy direction. The Philippine Information Agency (2025) reported that South Cotabato pursued a waste-to-energy initiative involving organic waste, biogas direction, and public-private partnership. This gives the province a strong environmental and resource recovery basis. In the checklist, it is also the only area rated "Yes" for potential buyers or off-takers because its evidence shows a clearer waste-to-resource pathway.

Bukidnon Province was considered because it has technical biomass evidence. Villar et al. (2016) conducted a GIS-based estimation of biomass availability and energy potential from corn residue in Bukidnon. This gives the province a strong basis for biomass resource and spatial assessment. Its inclusion is important because it shows how crop-residue data and GIS-based methods can support biomass screening at a wider area level.

The Davao del Norte/Tagum area was considered because Davao del Norte has biomass-related evidence connected to banana production. Agustin et al. (2024) used GIS and multi-criteria analysis to assess banana-based biomass power plant site suitability in Davao del Norte. This gives the area a clear biomass and renewable-energy basis. Tagum is included as a recognized urban and access reference within the area, not as the only biomass source.

Overall, Table 7 shows that all considered areas have valid reasons for inclusion in the analysis. Davao City is strong in organic agriculture governance, Cagayan de Oro City in agricultural logistics planning, South Cotabato in waste-to-resource direction, Bukidnon in GIS-based biomass assessment, and Davao del Norte in biomass power site suitability. However, Butuan City shows the broadest and most balanced evidence because it has documented multiple biomass streams, quantified biomass potential, collection and logistics concerns, agriculture and market-linkage background, digital agriculture precedent, and a long-term

renewable-energy plan. This makes Butuan City the strongest LGU choice based on the comparative scoping results.

2.1.1 Synthesis

Tables 6 and 7 show that the selection of Butuan City is not based on a single reason, such as having farms or producing agricultural waste. The deeper point is that a strong biomass study area must have both resource evidence and system needs. If an area only has biomass but no clear planning direction, the study becomes too general. If an area only has environmental programs but no documented residue sources, the biomass basis becomes weak. The strongest site is the one where biomass is present, the gaps are visible, and the local setting has enough structure to support further study.

Table 6 sets the standard for choosing a site. It shows that biomass scoping should look at the whole situation: the residue source, the location of supply, the possibility of recovery, the role of the LGU, the presence of users, and the availability of public evidence. This matters because biomass is not automatically useful just because it exists. It must be identified, located, verified, collected, and connected to a possible purpose. This is why the criteria are stronger when they are based on outside references such as Republic Act No. 9513 (2008), Republic Act No. 9003 (2001), Republic Act No. 7160 (1991), and biomass assessment studies, instead of being based on the proposed study itself.

Table 7 then shows the actual situation of the compared areas. The other LGUs have valid strengths, but most of them show a narrower type of evidence. Some are stronger in environmental programs, some in agricultural logistics, some in technical biomass studies, and some in renewable-energy site assessment. However, Butuan City shows a more complete situation because the evidence covers several important layers at the same time: documented biomass sources, varied residue types, collection concerns, logistics issues, agriculture support, digital or market-linkage background, and long-term energy planning.

The strongest point in favor of Butuan is that its evidence shows both potential and difficulty. Preferred Energy, Inc. and WWF Philippines (2025) documented rice, coconut, corn, and palm oil residues in Butuan, but they also reported that some residues are scattered, costly to collect, or affected by undeveloped logistics. This is

an important finding because it shows that the issue is not only the presence of biomass, but the lack of organized information and coordination around it. In research terms, this makes Butuan a stronger site because the problem is already visible and supported by evidence.

The Butuan City Energy Development Plan 2023–2050 also strengthens the selection because it connects the biomass discussion to a longer local direction. The city is not only being studied as an agricultural area; it is also a city with a long-term energy and sustainability plan. This gives the scope more weight because biomass is not treated as an isolated waste issue, but as part of a wider development concern involving resource recovery, renewable energy, and local planning.

Overall, the two tables show that Butuan City is the most defensible choice because it has the best balance of evidence, need, and local direction. Other areas may be strong in one or two parts, but Butuan shows the clearest combination of biomass availability, practical collection problems, governance support, sustainability direction, and available documentation. This makes the selection stronger because it is not based on convenience; it is based on a real situation where agricultural biomass exists, but its use still depends on better records, validation, aggregation, and coordination.

2.2 Hierarchy of Theoretical and Conceptual Foundations

The study uses several theories and concepts, but they do not have the same role. The main foundation is Design Science Research because the study produces a system artifact. The core problem theory is Information Quality and Data Quality Theory because the main issue is whether biomass records are complete, current, credible, and fit for use. The supporting concepts explain why the artifact must include stakeholder roles, source location, supply conditions, circular economy purpose, and software quality evaluation.

Table 8. Theory Hierarchy of AgriCycle

Level	Foundation	Role in the Study	Connection to CB-BAIM
Main foundation	Design Science Research	Explains why developing and evaluating a system artifact can produce research knowledge.	Guides problem identification, design, development, demonstration, and evaluation of AgriCycle.

Core problem theory	Information Quality and Data Quality Theory	Explains why biomass data must be fit for real use, not only present in a database.	Justifies validation, completeness, currentness, credibility, standard fields, and status labels.
Supporting lens	Socio-Technical Systems perspective	Explains why the system must match users, roles, rules, and workflow.	Links farmers, hubs, mills, buyers, LGU users, and the system administrator in one controlled process.
Supporting lens	Biomass Supply Chain Concept	Explains why actual availability depends on timing, storage, transport, handling, moisture, and aggregation.	Turns supply concerns into system fields such as source, period, status, storage, quality notes, and hub records.
Supporting lens	Spatial Decision Support / GIS	Explains why location is needed for local planning and coordination.	Supports maps of sources, mills, hubs, and collection points.
Supporting lens	Circular Economy	Explains the waste-to-value direction without claiming automatic income, energy, or emission impact.	Frames the system as an information layer before actual reuse, recovery, or conversion.
Evaluation basis	ISO/IEC 25010	Provides a software quality basis for system evaluation.	Guides evaluation of Functional Suitability and Usability.

2.2.1 Main Foundation: Design Science Research

Design Science Research is appropriate because AgriCycle develops and evaluates an information system artifact. Hevner et al. (2004) explain that design science produces knowledge through artifacts that respond to relevant problems. Peffers et al. (2007) describe a process that begins with problem identification and moves to objectives, design, development, demonstration, evaluation, and communication. This fits the study because the artifact is not made only as a programming output. It is made to express and test the CB BAIM Model.

In Chapter 3, this foundation will be used to organize the development process. The problem will be translated into objectives, system requirements, system design, system functions, and evaluation activities. The contribution will therefore come from both the working prototype and the model behind it.

2.2.2 Core Problem Theory: Information Quality and Data Quality Theory

Information Quality and Data Quality Theory directly supports the main claim of the study. Wang and Strong (1996) describe data quality as fitness for use. ISO/IEC 25012 also gives quality characteristics such as accuracy, completeness, consistency, currentness, credibility, and accessibility. These ideas fit AgriCycle because a biomass record is useful only when it can support a decision. A rough estimate without source, timing, location, status, and verification can create false confidence.

This is why the study treats validation as part of information quality, not just as an extra feature. A record becomes stronger when it shows who encoded it, what source it came from, where it is located, when it is available, how it is stored, whether it is already used, and who checked its status. Chapter 3 will use this theory by designing required fields, status labels, validation steps, reports, and audit logs.

2.2.3 Supporting Lens: Socio-Technical Systems Perspective

The socio-technical perspective is relevant because biomass availability cannot be managed by software alone. Farmers may know what residues remain after harvest. Rice mills may know actual husk generation and internal use. Cooperatives may know possible consolidation points. Buyers may know demand. LGU offices may need records for planning and monitoring. If these roles are not reflected in the system, the output may become a simple database with weak participation.

For this reason, AgriCycle uses controlled roles and a community-based validation process. The system does not require every farmer to manage a full online account. Instead, authorized local actors such as cooperative hubs, rice mills, institutional users, or LGU representatives may encode, update, or check records. This supports participation while reducing the risk of unreliable listings.

2.2.4 Supporting Lens: Biomass Supply Chain Concept

The Biomass Supply Chain Concept explains why biomass information must include more than type and quantity. Agricultural residues are often bulky, seasonal, scattered, and affected by moisture, collection, storage, labor, handling, and distance.

Recent supply chain studies also show that community-scale biomass planning depends on remaining residues, geospatial conditions, road networks, and local demand (Boonman et al., 2025).

In this study, those supply chain concerns are translated into simple information requirements. The system records source location, availability period, quantity, storage or quality notes, status, and aggregation details. Chapter 3 will not perform advanced route optimization or plant siting. Instead, it will build the information layer needed before more advanced planning can be done.

2.2.5 Supporting Lens: Spatial Decision Support and GIS-Based Decision Support

Spatial decision support explains why location must be part of biomass availability management. A residue source may have a large quantity, but it may still be difficult to use if it is far from a hub, scattered across many small sources, near poor roads, or already tied to another use. GIS helps users see where sources, mills, hubs, and collection points are located. In AgriCycle, GIS is used for local visibility and planning support, not for prediction or routing optimization.

2.2.6 Supporting Lens: Circular Economy

Circular Economy supports the waste-to-value direction of the study. It explains why agricultural residues should not automatically be treated as disposal materials when they may be used for composting, biochar, fuel, animal bedding, processing, or other suitable uses. At the same time, the study avoids overclaiming. It does not measure actual energy output, income increase, or emission reduction. Its contribution is the information layer that helps stakeholders identify, verify, and coordinate recoverable residues before any physical recovery happens.

2.2.7 Evaluation Basis: ISO/IEC 25010

ISO/IEC 25010 gives the software quality basis for evaluating the system. This study focuses on Functional Suitability and Usability. Functional Suitability checks whether the system functions are complete, correct, and useful for the intended

problem. Usability checks whether the intended users can understand and use the system with reasonable ease. These two criteria are enough for the system because the study does not claim a full production deployment.

2.3 Development of the CB-BAIM Model

The CB-BAIM Model is the formal research contribution of the study. It was developed from a recurring gap in biomass literature and related systems: many works estimate, map, trade, or plan biomass resources, but fewer provide a local workflow for checking whether a listed residue is practically available before aggregation or inquiry. The model answers this gap by treating availability as an information condition. A residue should pass through source identification, data capture, validation, information management, and coordination use before it is treated as usable decision information.

Figure 2. Community-Based Biomass Availability Information Management (CB BAIM) Model

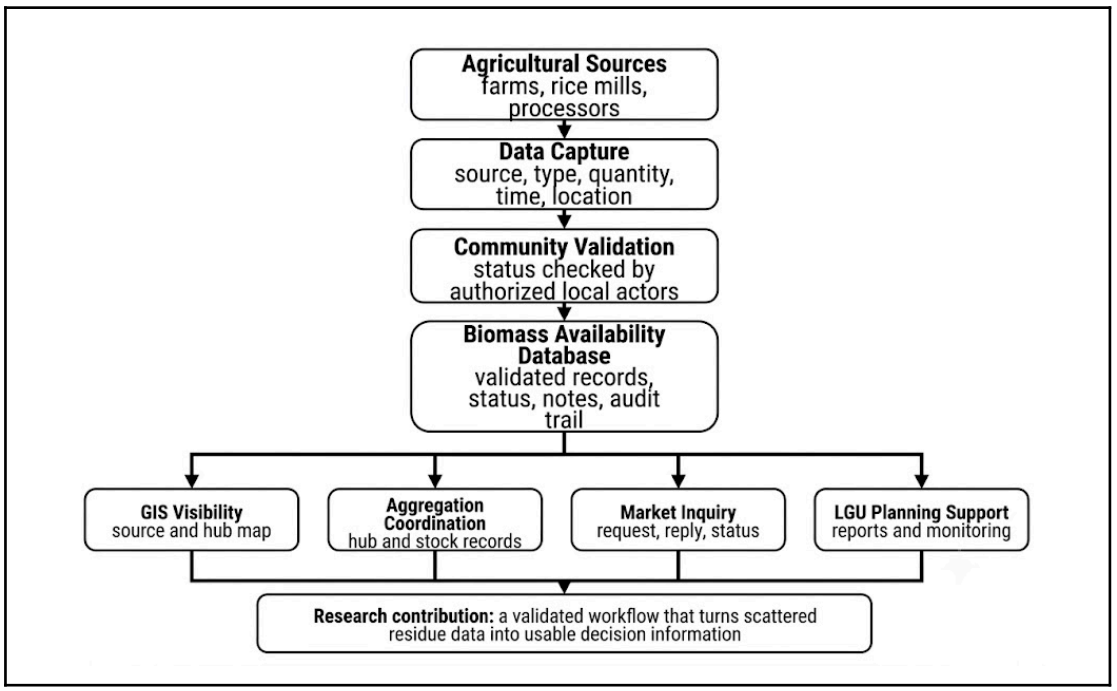


Figure 1 shows the logic of CB-BAIM. The process begins with local sources, moves to structured data capture, passes through community validation, and enters a biomass availability database. From that database, stakeholders can use GIS visibility, aggregation coordination, market inquiry, and LGU planning support. The value of

the model is not the number of modules. Its value is the controlled flow from scattered residue reports to checked decision information.

Table 9. CB-BAIM Model Logic and System Translation

CB-BAIM Stage	Research Logic	System Translation
Source identification	Biomass information begins by knowing where the data comes from.	Farms, rice mills, processors, cooperatives, hubs, and collection points are identified.
Data capture	Availability information needs standard records, not only informal messages or rough estimates.	Biomass type, quantity, source, location, period, contributor, and notes are encoded.
Community-based validation	Availability should be checked by people close to the source or authorized to verify local records.	Cooperative hubs, mills, LGU users, or institutional users check and update record status.
Information management	Validated records must be organized, mapped, updated, and traceable.	Inventory, GIS map, reports, dashboard, quality notes, status labels, and audit logs are provided.
Coordination use	Checked information becomes useful when stakeholders can act on it.	Aggregation coordination, inquiry handling, monitoring, and LGU planning support are enabled.

Table 9 shows that the system functions are not random features. Each function comes from a specific information problem. This strengthens the research contribution because AgriCycle is not defended as a simple listing platform. It is defended as a validated biomass availability information management workflow.

2.4 Philippine and Local Studies Related to Biomass Availability

Philippine and local studies are important because the study is based in Butuan City and involves local agricultural residues, LGU participation, and community-based information. International literature gives useful theory and methods, but the Philippine context shows the actual residue practices, collection problems, and digital agriculture conditions that make the proposed system relevant.

Elepaño et al. (2020) examined major agricultural residues in the Philippines and noted that rice straw, corn cobs, coconut husks, rice hulls, sugarcane bagasse, corn stover, and coconut shells are common residues from major crops. The study also reported that many residues are burned, left in the field, used as household fuel, or affected by losses during collection and transport. This supports AgriCycle because it

shows that residue presence is different from residue availability. A residue may exist, but its actual condition, current use, and recoverability still need checking.

Rice straw management literature further shows the local difficulty of collection. PHilMech and the Department of Agriculture describe rice straw as commonly treated as waste, with field collection described as labor-intensive and difficult. They identify possible good practices such as soil incorporation, baling, mushroom substrate, silage, composting, fuel use, and biochar. This supports the study because the system should not assume that rice straw is ready for use only because it appears after harvest.

Coconut residue studies also support the need for local validation. Pogosa et al. (2021) assessed coconut husk availability and utilization in major coconut-producing provinces in the Philippines. Their study used surveys and interviews with coconut farmers and showed that husk production, density, and utilization differ across provinces. They also noted that large volumes of husks may remain as farm waste or be burned, while some are used as firewood or for fiber and peat-based products. This strengthens the need to record present use and remaining availability before coconut residues are listed for coordination.

Mindanao-based studies also show that biomass assessment often focuses on estimation and mapping. Villar et al. (2019) used survey questionnaires, government agriculture reports, and ArcGIS to estimate corn residue biomass and energy potential in Bukidnon. Golez et al. (2022) assessed the biomass energy potential of five major crops in Mindanao using GIS and remote sensing. Galang et al. (2021) used GIS and AHP to analyze suitable locations for sugarcane residue production in Medellin, Cebu. These studies support GIS-based resource planning, but they also show the gap that AgriCycle addresses: local maps and estimates still need a system workflow that updates and validates practical availability at the community level.

Philippine digital agriculture programs show that information systems can support farmers and agricultural users, but they also show that digital readiness must be considered. The Agricultural Training Institute has programs such as the Digital Farmers Program and e Extension for agriculture and fisheries, which help farmers access information online, use digital tools, and connect with market opportunities. The Department of Agriculture also launched eKADIWA as a digital market linkage

platform for agricultural and fishery products. These programs support the use of digital systems in agriculture, but AgriCycle differs because it focuses on biomass residues, validation, GIS-based availability records, aggregation coordination, and LGU planning support.

Table 10. Philippine and Local Literature

Philippine / Local Source	Main Focus	Relevance to AgriCycle
Elepaño et al. (2020)	Major agricultural residues in the Philippines and their bioenergy potential.	Shows that residues are common but may be burned, left in fields, used locally, or reduced by collection and transport losses.
PHilMech / DA (2024)	Rice straw management in the Philippines.	Shows that rice straw collection is labor intensive and that several recovery options require organized collection and local support.
Pogosa et al. (2021)	Coconut husk availability and utilization in major coconut provinces.	Shows the need to assess actual production, present use, and remaining availability of coconut husks.
Villar et al. (2019)	GIS-based corn residue biomass and energy potential in Bukidnon.	Supports Mindanao biomass mapping but also shows that mapping must be linked with local availability records.
Golez et al. (2022)	Biomass energy potential of five major crops in Mindanao.	Supports the Mindanao biomass context and the use of GIS for resource concentration.
Galang et al. (2021)	GIS-AHP analysis for sugarcane residue location in Cebu.	Shows the role of location, topography, cultivation area, and accessibility in biomass planning.
ATI Digital Farmers Program and e Extension	Digital literacy and ICT-based agricultural extension.	Supports the digital agriculture context, but AgriCycle focuses on verified biomass records rather than general farm training or food marketing.
DA eKADIWA	Digital market linkage for agricultural and fishery products.	Shows market linkage precedent, but not residue validation, GIS inventory, or LGU biomass planning.

The local literature strengthens the research gap. Philippine studies already show that agricultural residues exist and may have value. They also show that residues are affected by burning, field practices, collection effort, transport, storage, and present use. Therefore, the missing part is not proof that residues exist. The missing part is a local information system workflow that can check which residues remain practically available and useful for coordination.

2.5 Butuan City Biomass and Energy Planning Context

The Butuan City Energy Development Plan 2023-2050 identifies biomass as one possible renewable energy resource and includes residues from rice, coconut, corn, and palm oil in the city resource assessment. However, the biomass pre-feasibility study also shows that theoretical estimates should be treated carefully because some residues are already used, scattered, difficult to collect, or not enough to support a stable 1 MW biomass power supply under current conditions (City Government of Butuan, 2023; Preferred Energy, Inc. & WWF Philippines, 2025).

This local finding is important because it prevents the study from overclaiming. AgriCycle does not assume that the city has unlimited biomass supply. Instead, it responds to a more realistic problem: the city needs updated and validated information to separate estimated biomass potential from practical biomass availability. This also explains why the system remains useful even if a biomass power plant is not immediately feasible. Waste-to-value planning still needs reliable records on available residues, possible hubs, current use, buyers, and local status changes.

2.6 Biomass Information Quality and Data Management

Biomass information quality is central to the study. A record is weak if it only states a quantity. It becomes useful when it includes source, location, period of availability, present use, storage condition, quality concern, validation status, and update history. For a cooperative, useful information means knowing what can be consolidated. For a buyer, it means knowing whether a listing is worth an inquiry. For an LGU office, it means having records that can support monitoring, reports, and future waste-to-value planning.

This section connects directly to Chapter 3. Because the literature shows that data quality depends on fitness for use, the methodology will include required data fields, validation status, role-based access, audit logs, and user evaluation. The system will not only display biomass records. It will structure and check them so the records can be used with less uncertainty.

2.7 Stakeholder Ecosystem and Community-Based Validation

A biomass availability system depends on stakeholder interaction. Information is not stored in one place. Farmers may know field residues, rice mills may know husk generation, cooperatives may know available hubs, buyers may know demand, and LGU offices may need data for planning. This makes the problem socio-technical because the system must connect people, records, rules, and technology.

Community-based validation is used because direct public listing may create unreliable records. If any user can post biomass quantities without checking, buyers may see duplicate, outdated, or unavailable supply. At the same time, requiring every farmer to manage a digital account may not be realistic because of device access, digital skill, and time constraints. The study therefore uses authorized local actors to encode, verify, and update records while still allowing farmer information to pass through hubs, mills, cooperatives, or LGU-supported channels.

Table 11. Stakeholder Ecosystem for Community-Based Biomass Availability Management

Stakeholder	Information Position	Role in CB-BAIM
Smallholder farmers	Know residue generation after farming activities but may not use the system directly.	Provide information through cooperatives, hubs, mills, or authorized users.
Rice mills and processors	Know actual residue production, internal use, and remaining volume.	Provide or update records on rice husk, processing residues, status, and availability.
Cooperatives or hubs	Can consolidate records from farmers and processors.	Encode, validate, update, and monitor aggregation records.
LGU offices	Need data for agriculture, environment, planning, energy, and waste-to-value programs.	Use dashboards, maps, reports, and logs for monitoring and planning.
Institutional buyers or users	Need reliable information before making inquiries.	View validated listings and submit inquiries without assuming that a legal transaction is completed.

The stakeholder structure also protects the system scope. AgriCycle is not a procurement platform and does not award contracts. Its role is to create a controlled space where checked biomass information can be used for aggregation, inquiry, reporting, and planning.

2.8 GIS-Based Biomass Mapping and Local Planning

GIS-based biomass literature supports the spatial part of the study. Many studies use GIS to estimate potential supply, identify resource concentrations, or screen suitable locations for bioenergy projects. These outputs are useful, but they can still be incomplete for local coordination if the records do not show present use, validation status, availability period, or update history.

In AgriCycle, GIS is used in a simple and local way. It helps users view biomass sources, rice mills, hubs, and collection points. The map is not meant to prove that all displayed residues are ready for collection. Instead, it connects location with validated availability information. Chapter 3 will therefore include GIS visualization as part of the prototype, but it will remain tied to record status and validation fields.

2.9 Biomass Supply Chain, Aggregation, and Practical Availability

Biomass supply chain studies show why agricultural residues should not be treated as available only because they exist. Residues must be collected, stored, handled, transported, aggregated, and matched with use. These steps are affected by season, moisture, bulk, low density, labor, storage space, distance, and local demand. These conditions explain why the study uses the term practical biomass availability.

In this study, practical availability means that the residue is not only present but also recorded with enough information for a decision. A buyer should not only see a quantity; the buyer should also know the location, time, status, and whether the record has been checked. A cooperative should not only know that residue exists; it should know whether it can be consolidated through a hub. An LGU should not only have city-level estimates; it should have updated records that can support planning and monitoring.

This section also guides Chapter 3. The system will include inventory, aggregation records, status labels, and reports because these are needed to reflect real supply conditions. The study will not solve all physical logistics problems, but it will organize the information needed before logistics decisions can be made.

2.10 Market Inquiry and Coordination Scope

The term market inquiry is used carefully in this study. AgriCycle is not a full e-commerce, bidding, payment, delivery booking, or legal procurement system. It is an inquiry and coordination space. The system allows interested users to view validated biomass information and submit inquiries, but any actual sale, payment, contract, transport arrangement, or procurement process remains outside the system scope.

This limited scope is important because the real contribution of the study is availability information management. Market inquiry is only one use of the validated records. The system first asks whether the biomass information is complete, current, checked, and useful. Only after that can the record support inquiry or coordination. This avoids the mistake of making the study appear to be a biomass trading business when its stronger identity is a validated biomass information system.

Chapter 3 will follow this scope by evaluating inquiry-related functions only as coordination features. The evaluation will not measure actual sales, income, contracts, procurement awards, or delivery performance.

2.11 Recent Journals and Studies Related to the Study

The journal studies reviewed for this chapter support the same point from different angles. Some studies focus on GIS mapping, some on biomass supply chains, some on waste to-value recovery, and some on digital or community-based data systems. Together, they show that biomass planning already has strong support in literature. However, they also show the need for a local workflow that checks practical availability before records are used for aggregation, inquiry, or LGU planning.

Table 12. Recent Journals and Studies

Study	Main Focus	Connection to AgriCycle
Jusakulvijit et al. (2022, Thailand)	GIS-based multi-criteria and logistics assessment for agricultural residue valorization.	Supports GIS and logistics considerations for biomass source mapping and aggregation planning
Chidozie et al. (2023, international review)	Systematic review of residual agroforestry	Supports the need to study collection, transport, storage, and supply chain

	biomass supply chain simulation.	behavior for biomass residues
Duque-Acevedo et al. (2023, Spain/Czech Republic)	Review of agricultural waste biomass generation, use, and regulation in circular production models.	Supports the waste-to-value context while showing that policy and actual use must be considered.
Valipour et al. (2024, international review)	Review of bio-hubs in biomass supply chains.	Supports the hub-based concept for consolidating biomass information and supply.
Roudneshin et al. (2024, international review)	Review of agricultural waste supply chain optimization for bioenergy	Supports the inclusion of transport, storage, cost, and supply coordination concerns.
Chidozie et al. (2024, international)	Methodological framework for residual biomass supply chain simulation.	Supports the idea that biomass supply chains need structured modeling and reliable input data.
Naeimi et al. (2025, Iran)	GIS mapping and neural network method for crop-residue biomass energy potential.	Supports the use of spatial data to estimate and visualize biomass sources.
Bastos et al. (2025, Portugal)	Digitalization of residual biomass supply chain.	Supports the role of digital systems in organizing biomass logistics and decision data.
Masi et al. (2025, Europe)	Digital platform for short food supply chain monitoring and value creation.	Supports the use of local digital platforms for partner records, distribution data, and decision support.
Namaswa et al. (2026, Kenya)	Spatial distribution and availability of residues for biochar feedstock.	Supports the distinction between residue presence, surplus availability, and practical use.
Ukoba et al. (2024, Nigeria)	Multicriteria GIS-based assessment of biomass energy potential.	Supports spatial screening and confirms that GIS studies often focus on suitability and potential.
Bastos et al. (2025, Portugal)	Web app-based information model for residual biomass recovery.	Supports the idea that residual biomass recovery needs information flow, stakeholder connection, transparency, and logistics support.
Ghalandarzadeh et al. (2025, Sweden)	Community-based business models for agriculture and forestry data ecosystems.	Supports the community-based data ecosystem logic and shows both the value and barriers of multi-stakeholder data models.
Boonman et al. (2025, Thailand)	Sustainable biomass supply chain design for community-scale biomass power plants.	Supports the importance of remaining residues, geospatial criteria, road networks,

Table 12 restores the journal-based review. The table shows that existing journals already support GIS mapping, supply chain analysis, digital platforms, and

biomass recovery. The remaining gap is the limited treatment of community-based validation of practical biomass availability before aggregation and inquiry. This is why CB-BAIM is positioned as an information management contribution rather than a simple marketplace feature.

The recent journal literature shows that biomass waste-to-value management depends not only on the amount of agricultural residue present, but also on the quality of information used to identify, locate, validate, and coordinate that biomass. Studies on GIS-based biomass assessment show that location data is important because agricultural residues are often distributed across farms, mills, and collection points. Jusakulvijit et al. (2022) and Naeimi et al. (2025) support this point by showing how GIS can be used to assess biomass potential from agricultural residues. This is directly related to AgriCycle because the proposed system uses GIS mapping to show biomass source locations, rice mills, cooperative hubs, and collection points. However, AgriCycle goes beyond mapping because it also records whether biomass is available, already used, reserved, unsuitable, or still pending validation.

The reviewed supply chain studies also show that biomass availability must be understood in relation to collection, storage, transport, timing, and coordination. Chidozie et al. (2023), Chidozie et al. (2024), Roudneshin and Sosa (2024), Boonman et al. (2025), and Valipour et al. (2024) explain that biomass supply chains require organized data and coordination among different actors before residues can be used effectively. This supports the cooperative-based aggregation feature of AgriCycle. In the proposed system, cooperatives and authorized users are not only data encoders; they also help validate and consolidate biomass information before it is used for reports or buyer coordination. This is important because biomass may exist in the area but may not be practically available due to present use, poor storage, high moisture, distance, or seasonal collection limits.

The studies on circular agricultural waste management and digital platforms further support the waste-to-value and marketplace direction of the system. Duque-Acevedo et al. (2023) show that agricultural waste biomass can support circular production when actual use, policy, and management are connected. Bastos et al. (2025) highlight the importance of digitalization in improving residual biomass supply chains, while Masi et al. (2025) show that digital platforms can help connect producers and users through better information exchange. Namaswa et al. (2026) also

emphasizes that residue presence should be separated from practical availability because not all biomass can be collected or used economically. These findings support the main contribution of AgriCycle: a local information system that validates biomass availability, organizes inventory records, maps sources, supports cooperative aggregation, and manages buyer inquiries. The reviewed studies provide strong support for the system, but they do not fully address the specific need for a community-based biomass availability information management system for Butuan City.

2.12 Review of Related Systems

The reviewed systems show that many existing platforms already support parts of AgriCycle's direction. Some systems are strong in GIS resource mapping, some are strong in agricultural market linkage, and others are strong in biomass supply chain planning. However, most do not combine practical biomass availability validation, cooperative-based aggregation, GIS mapping, buyer inquiry coordination, and LGU-ready reports in one local system.

Table 13. Review of Related Systems

No.	Related System	Main Features	Relevance to AgriCycle	Gap Addressed by AgriCycle
1	AgriMax Online Marketplace (2025, Europe)	Online platform connecting crop producers, food processors, and agricultural cooperatives with biorefineries for agricultural waste and raw materials.	Related to biomass listing and waste-to-value market linkage.	Focused on biorefinery feedstock selling; does not provide barangay-level validation, cooperative workflow, LGU reports, or Butuan-specific biomass inventory
2	BALTPOOL Biomass Exchange and Traceability Tool (2021, Europe)	Biomass exchange with traceability tools for biomass origin, quantity, raw material, and trading transparency.	Related to biomass market transparency and origin tracking.	Designed for organized biomass trading; not a community-based agricultural residue availability system for farms, mills, and cooperatives
3	BioenergyKDF (2024, United States)	Central bioenergy data portal with resource maps, data access, and geospatial decision support.	Related to resource mapping and geospatial data sharing.	National-level resource portal; does not validate local practical availability or support cooperative aggregation and buyer inquiries in Butuan
4	BiomassConnect Information Hub (2022/2026, United Kingdom)	Knowledge-sharing and demonstration platform for land-based biomass feedstock production and best practices.	Related to biomass information sharing and feedstock knowledge.	Information and demonstration hub; not an inventory, validation, aggregation, or marketplace inquiry system
5	BIOPLAT-EU WebGIS Tool (2022, European Union)	WebGIS platform for sustainable bioenergy planning using marginal, underutilized, and contaminated lands.	Related to GIS-based screening and bioenergy planning.	Focuses on land suitability and investment planning; does not validate agricultural residue availability at the community level.
6	BioRural Toolkit (2022/2025, European Union)	Online toolkit that integrates information on small-scale bio-based solutions, rural actors, research, policy, and financing.	Related to rural bioeconomy information access and stakeholder learning.	Knowledge toolkit; does not manage live biomass records, validation statuses, inventory, or buyer inquiries.
7	BSWM Soil GIS and Information Systems (2024, Philippines)	Agricultural GIS and soil-water information systems for spatial data, land resources, hazards, and agricultural planning.	Supports the importance of location-based agricultural information.	Focused on soil and land resources; not a biomass availability, aggregation, or marketplace system.
8	Butuan AgriBOOST / AgriMart (2024, Philippines)	Local hub and mobile platform for vegetable orders, demand consolidation, farm delivery schedules, and buyer-farmer linkage.	Highly relevant because it shows a Butuan hub-and-app model for agricultural coordination.	Focused on vegetables and market orders; not designed for biomass residues, present use validation, storage condition, or residue quality notes.

9	DA-PRDP Applied Geotagging Tool / GeoCamera (2020/2026, Philippines)	Geotagging and geovideo tools for project monitoring, location verification, and transparency in rural development projects.	Related to location validation and transparent records.	Tracks projects and infrastructure activities; not biomass residue availability or aggregation records.
10	eKADIWA Digital Marketing Platform (2020, Philippines)	Digital marketing application linking farm producers to consumers and supporting direct market access.	Related to digital market linkage between producers and buyers.	Focused on food products and delivery; does not validate biomass residues or manage cooperative biomass inventory.
11	ENABLING Biomass Trade Platform (2020, European Union)	Online venue where biomass producers and processors can offer and request biomass residues, by-products, and services.	Related to biomass listings and buyer-supplier matching.	Provides exchange functions but not local validation, GIS inventory, cooperative workflow, or LGU reporting for Butuan.
12	MainstreamBIO Digital Toolkit (2024/2026, European Union)	Toolkit for matching available biomass and market trends with bio-based technologies and regional value-chain information.	Related to matching biomass with biobased opportunities and regional planning.	Designed for regional bioeconomy support; not a live barangay-level inventory and marketplace inquiry system.
13	Philippine Rice Information System or PRiSM (2024/2025, Philippines)	Satellite-based rice monitoring system that provides rice area, yield, and risk data.	Related to rice production data, seasonal monitoring, and GIS-based agricultural information.	Tracks rice production conditions; does not track rice straw, rice husks, present use, residue availability, or buyer coordination.
14	Rice Crop Manager Advisory Service or RCMAS 4.0 (2021, Philippines)	Digital agriculture service for rice crop and farm data management and farmer advisories.	Supports assisted digital agriculture and structured farm information.	Focused on crop recommendations; not biomass residue availability, validation, aggregation, or marketplace coordination.
15	Rice Straw Biogas Hub / Dayami Hub (2025, Philippines)	Rice straw bioenergy hub model that demonstrates rice straw collection and conversion into energy and other value-added products.	Supports the cooperative and hub-based concept for agricultural waste-to-value management.	Physical bioenergy initiative; not a digital information management system with GIS, inventory, validation, and buyer inquiry modules.

2.13 Comparative Feature Checklist of Related Systems

The checklist below compares the related systems based on the features required by the present study. “Yes” means the available documentation shows that the system supports the feature. “Partial” means the feature is related but not complete or not directly focused on biomass availability. “No” means the feature is not shown in the available documentation. The comparison shows that many systems are useful, but most of them support only one part of the problem. AgriCycle combines practical biomass availability validation, GIS mapping, cooperative aggregation, inventory records, marketplace inquiry coordination, and reports for local planning.

Table 14. Feature Checklist Comparing Related Systems

No.	System	Availability Validation	GIS/ Location Data	Use/Storage/ Quality Data	Aggregation Workflow	Marketplace/ Inquiry	Reports/ Planning	Interpretation
1	AgriCycle	Yes	Yes	Yes	Yes	Yes	Yes	Designed to combine these features for Butuan City.
2	AgriMax Online Marketplace	Partial	No	Partial	Partial	Yes	Partial	Strong in the agricultural waste marketplace, but not local validation or LGU reporting.
3	BALTPOOL Biomass Exchange and Traceability Tool	Partial	Partial	Partial	Partial	Yes	Partial	Strong in trading and traceability, but not community-based farm and cooperative validation.
4	BioenergyKDF	Partial	Yes	No	No	No	Yes	Strong in resource mapping, but not local availability validation or marketplace inquiry.
5	BiomassConnect Information Hub	Partial	Partial	Partial	No	No	Partial	Strong in information sharing, but not inventory and transaction coordination.
6	BIOPLAT-EU WebGIS Tool	Partial	Yes	Partial	No	No	Yes	Strong in WebGIS screening, but not residue validation and cooperative

								workflow.
7	BioRural Toolkit	Partial	Partial	Partial	Partial	No	Partial	Supports rural bioeconomy learning, but not live biomass records and buyer inquiries
8	BSWM Soil GIS and Information Systems	No	Yes	No	No	No	Yes	Strong in agricultural spatial data, but not biomass residue coordination
9	Butuan AgriBOOST / AgriMart	Partial	Partial	No	Yes	Yes	Partial	Strong local hub model, but focused on vegetables rather than biomass residues
10	DA-PRDP Applied Geotagging Tool / GeoCamera	No	Yes	No	No	No	Yes	Strong in location verification, but focused on project monitoring.
11	eKADIWA Digital Marketing Platform	No	No	No	Partial	Yes	No	Market linkage for food products, not biomass residue availability.
12	ENABLING Biomass Trade Platform	Partial	No	Partial	Partial	Yes	No	Biomass exchange exists, but local validation and GIS inventory are not central.
13	MainstreamBIO Digital Toolkit	Partial	Partial	Partial	Partial	Partial	Yes	Supports matching and regional planning, but not local live inventory.
14	Philippine Rice Information System (PRiSM)	No	Yes	No	No	No	Yes	Rice monitoring system, but not residue availability and marketplace coordination.
15	Rice Crop Manager Advisory Service (RCMAS 4.0)	No	Partial	No	No	No	No	Crop advisory service, not biomass residue aggregation.
16	Rice Straw Biogas Hub / Dayami Hub	Partial	Partial	Partial	Yes	Partial	Partial	Physical waste-to-value hub, but not a digital availability management platform.

2.14 System Development and Evaluation Link to Chapter 3

The literature leads directly to the system requirements. Since biomass information must be checked, the system needs validation status and role-based access. Since supply conditions affect availability, the system needs fields for source, quantity, period, storage or quality notes, and aggregation status. Since location affects coordination, the system needs GIS visibility. Since stakeholders have different roles, the system needs controlled user accounts. Since the artifact must be evaluated, the study uses ISO/IEC 25010 for Functional Suitability and Usability.

Table 15. Literature-to-Methodology Bridge

Literature Finding	Design Response in Chapter 3	Expected Evaluation Focus
Biomass estimates do not automatically show practical availability.	Create records with source, status, location, period, and validation fields.	Functional completeness and correctness of data capture.
Community actors hold important local information.	Use role-based accounts for hubs, LGU users, institutional users, and administrators.	Usability and clarity of user roles.
GIS supports planning but must be tied to record status.	Provide map-based visibility of sources, mills, hubs, and collection points.	Usefulness and clarity of GIS functions.
Aggregation requires updated stock and hub information.	Include inventory and aggregation coordination records.	Suitability of functions for cooperative or hub use.
Inquiry should not be confused with payment or procurement.	Limit market function to inquiry, reply, and status monitoring.	Clarity of scope and ease of inquiry use.
System quality must be tested with users.	Evaluate Functional Suitability and Usability using ISO/IEC 25010-based instruments.	User ratings and comments on system functions and interface.

Table 15 makes the connection between Chapter 2 and Chapter 3 clear. The review does not only describe related literature. It explains why specific system functions and evaluation criteria are needed.

2.15 Synthesis of the Review and Research Gap

The review shows that the study has a clear information-system contribution. Biomass studies already estimate resources, map potential supply, analyze logistics, and discuss waste to-value uses. Digital agriculture systems already support

information access and market linkage in different forms. However, the reviewed literature and systems provide limited mechanisms for validating practical biomass availability at the community level before aggregation, inquiry, and LGU planning use.

The research contribution of AgriCycle is the CB-BAIM Model. The model explains that agricultural residues become useful for waste-to-value coordination only when their availability is captured through standard fields, checked by authorized local actors, connected to location, organized into a database, and made usable for aggregation, inquiry, and reports. It is enough to move the study beyond ordinary system development because it proposes a clear workflow for turning scattered residue data into usable decision information.

The final research gap can be stated as follows: there is limited literature and system evidence on community-based information models that validate practical biomass availability before aggregation and market inquiry in a local LGU setting. AgriCycle addresses this gap by providing a Butuan-specific system and model for capturing, checking, mapping, consolidating, and sharing biomass availability information for agricultural waste-to-value management.

This gap also defines the scope of Chapter 3. The methodology will focus on the design, development, and evaluation of the prototype as an information system. It will test whether the system functions fit the CB-BAIM workflow and whether intended users can use the system clearly. It will not claim to measure actual biomass conversion, energy output, income generation, emission reduction, legal procurement completion, or full logistics performance.

CHAPTER 3

METHODOLOGY

This chapter presents the methodology of the study titled AgriCycle: A Community-Based Biomass Availability Information Management System with Aggregation and Inquiry-Based Market Linkage Coordination for Agricultural Waste-to-Value Management. It explains how the researchers will examine the biomass information problem, develop the system, and evaluate the finished system.

The study is centered on the Community-Based Biomass Availability Information Management (CB-BAIM) Model. The model treats biomass availability as an information management concern. Biomass should not be treated as available only because it appears in an estimate. It must be identified, recorded, checked, mapped, classified, consolidated, and shared with the right users before it can support planning or coordination.

The methodology follows a descriptive-developmental design guided by Design Science Research. Design Science Research is used because the study develops and evaluates a system artifact that responds to a real local problem (Hevner et al., 2004; Peffers et al., 2007). Agile methodology guides the system development because the system will be built through planned iterations, checking, feedback, and revision. Agile is applied in a simple academic form so the work remains manageable for a student project (Beck et al., 2001; Schwaber & Sutherland, 2020).

AgriCycle focuses on biomass records, community-based validation, GIS supported visibility, aggregation coordination, marketplace inquiry, reports, and audit logs. It will not include digital payment, bidding, procurement approval, legal contracts, route optimization, laboratory testing, or construction of a biomass facility. These limits keep the study within biomass availability information management.

3.1 Research Design

This study will use a descriptive-developmental research design. The descriptive part will examine current problems in biomass data collection, record keeping, validation, mapping, reporting, and stakeholder coordination in Butuan City.

It will also identify what information users need before biomass can be treated as practically available.

The developmental part will cover the design and development of AgriCycle. Findings from the descriptive phase will be converted into system requirements, process models, user roles, data fields, database structure, interface screens, and system functions.

Design Science Research will serve as the research approach. Under this approach, the system is not treated only as a programming output. It is treated as a research artifact that applies the CB-BAIM Model to a local information problem. The artifact will be designed, demonstrated, evaluated, and documented.

Agile methodology will guide the actual system building. The system will be developed in short iterations. Each iteration will include planning, design, coding, internal testing, feedback recording, and revision. This approach is suitable because the system has connected parts that must be checked before final user evaluation.

The study will use both qualitative and quantitative data. Qualitative data will come from document review, needs assessment, key informant interviews, process notes, system review notes, and open-ended comments. Quantitative data will come from the system evaluation questionnaire using a five-point rating scale.

Table 16. Research Questions, Methodology, Data Sources, and Outputs

Research Question	**Methodological Approach**	**Data Sources or Participants**	**Expected Output**
1. Examine the core problems affecting biomass availability management.	Document review, needs assessment, key informant interview, and process observation.	BEDP, biomass pre feasibility report, cooperative or hub personnel, rice mill personnel, LGU representatives, institutional users, and buyers when available.	Problem analysis, current process description, data gaps, stakeholder needs, and CB-BAIM input factors.
2. Design and develop AgriCycle based on the CB-BAIM Model.	Requirements analysis, BPMN, UML, ERD, system architecture, Agile development iterations, and internal review.	Needs assessment results, interview findings, process notes, requirements matrix, user role descriptions, and sprint review notes.	Developed prototype, As-Is and Proposed BPMN, Use Case Diagram, ERD, system architecture, data fields, validation workflow, GIS mapping, aggregation, inquiry, reports, and audit logs.

3. Evaluate the developed system in terms of Functional Suitability and Usability.	Internal testing, task based user testing, evaluation questionnaire, and open-ended feedback analysis.	Available qualified participants from the identified user groups and technical evaluators.	Evaluation ratings, weighted mean, standard deviation, interpreted results, user comments, and final revision notes.

3.2 Research Locale and Study Context

The study will be conducted in Butuan City. The city is used as the locale because its biomass issue is already supported by local planning documents and pre-feasibility evidence. The Butuan City Energy Development Plan 2023-2050 identifies biomass as one possible renewable energy resource for the city (City Government of Butuan, 2023). The biomass pre feasibility findings also show that theoretical biomass potential is different from collectable and usable supply because some residues are already used, scattered, seasonal, or affected by storage and transport conditions (Preferred Energy, Inc. & WWF Philippines, 2025).

The study context is focused on information management. It does not aim to build a biomass plant or run a physical collection program. It aims to help users record, check, map, consolidate, and share biomass availability information in a more organized way. This direction is consistent with the use of digital technologies in agriculture and rural information systems (Trendov et al., 2019).

3.3 Respondents and Participants of the Study

The study will involve available qualified participants who have direct knowledge of biomass records, agricultural residues, rice milling, cooperative aggregation, LGU planning, institutional biomass use, buyer coordination, or system evaluation. Participants will not be selected randomly because the needed information is specific to the study context. The respondents will be grouped according to their role. One group will help identify the present problems and user needs. Another group will help test and evaluate the system. Some participants may belong to both groups if they are involved in the current process and are also available during system evaluation.

The exact number of participants will depend on actual access, permission, and availability of the identified organizations. The researchers will request

permission from concerned offices, cooperatives, rice mills, institutions, or other relevant groups before gathering data. If access to a certain group is limited, the study will report the actual number reached and explain the limitation clearly.

For the needs assessment, the researchers will attempt to gather data from qualified key informants from the identified user groups. For system evaluation, the study will involve available qualified participants representing the intended users and technical evaluators. The purpose is not to generalize to all residents of Butuan City. The purpose is to gather informed data from people who understand the biomass information and coordination process.

Table 17. Participant Groups and Their Role in the Study

Participant Group	Reason for Inclusion	Main Role in the Study
Cooperative hub managers or aggregation personnel	They may handle encoding, checking, consolidation, storage notes, and updating of biomass records.	Needs assessment, requirements review, and system evaluation when available.
Rice mill personnel or agricultural processors	They can explain residue generation, internal use, storage, quality concerns, and remaining availability.	Needs assessment, validation workflow review, and system evaluation when available.
LGU representatives and institutional stakeholders	They may use reports, maps, and summaries for agriculture, environment, planning, energy, and BEDP monitoring.	Needs assessment, planning information review, and system evaluation when available.
Institutional buyers or biomass users	They may view validated biomass listings and send inquiries for composting, biochar, research, energy, or other waste-to-value uses.	Needs assessment and marketplace inquiry evaluation when available.
Evaluators (They can be the same group in above)	They can review the system from an information system or software development perspective.	Functional Suitability, and Usability.

3.4 Sampling Technique

Purposive sampling will be used. This technique is appropriate because the study needs respondents who can provide specific information about biomass records, rice milling, cooperative or hub operations, LGU planning, institutional biomass use, buyer inquiry, or system evaluation. Participants must be at least 18 years old and willing to give informed consent.

For Objective 1, participants must be able to answer questions about current biomass data collection, current use, storage, quality, location, validation, reporting, or stakeholder coordination. For Objective 2, selected participants may review the requirements, process flow, or system screens during development review. For Objective 3, participants must be able to use the system during testing or review it through a guided demonstration. Participants who are not connected to the study context or who do not give consent will not be included.

3.5 Research Instruments

The study will use instruments that match the three research questions. These instruments will help the researchers examine the current problem, convert findings into system requirements, and evaluate the developed system. Before actual use, the adviser and selected validators will review the instruments for clarity, relevance, and coverage of the required data.

Table 18. Research Instruments and Their Purpose

Instrument	Purpose
Document Review Matrix	Reviews the BEDP, biomass pre-feasibility findings, related studies, related systems, and available local records.
Stakeholder Needs Assessment Questionnaire	Identifies present data practices, record gaps, availability problems, storage concerns, quality notes, reporting needs, and coordination issues.
Key Informant Interview Guide	Gather deeper explanations from LGU users, cooperative personnel, rice mill personnel, buyers, and institutional users.
System Requirements Matrix	Converts identified problems into system functions, data fields, user roles, rules, and workflow requirements.
BPMN and UML Design Checklist	Checks whether the As-Is BPMN, Proposed BPMN, Use Case Diagram, ERD, and System Architecture fit the research scope and system design.
Agile Sprint Review Checklist	Checks whether the planned functions for each iteration were completed, tested, reviewed, and revised.
Internal Test Case Checklist	Checks common user actions before the system is shown to respondents.
ISO/IEC 25010-Based Evaluation Questionnaire	Evaluates Functional Suitability and Usability through a five-point rating scale.
Open-Ended Feedback Form	Records comments, observed issues, recommended improvements, and user experience concerns after testing.

3.6 CB-BAIM Operationalization in the Methodology

The CB-BAIM Model will be operationalized by translating each model component into research activities and system functions. This shows that the model is not only mentioned in the study. It is used as the basis for data gathering, requirements analysis, system development, and evaluation.

Table 19. Operationalization of the CB-BAIM Model in the Methodology

CB-BAIM Component	Methodology Activity	System Translation
Source Identification	Document review, interviews, and needs assessment will identify farms, rice mills, cooperatives, hubs, processors, buyers, and LGU users.	User roles, source profiles, biomass source records, and stakeholder categories.
Data Capture	Needs assessment will identify what data are currently recorded and what data are missing.	Standard fields for biomass type, quantity, location, contributor, current use, availability period, storage condition, and quality notes.
Community-Based Validation	Interviews and process review will identify who can check biomass records close to the source.	Validation roles, validation status, validation notes, and status change records.
Availability Classification	The researchers will identify useful status categories for biomass records.	Available, unavailable, reserved, already used, unsuitable, pending validation, or other approved categories.
Spatial Visibility	The study will determine which sources, mills, hubs, and collection points should be mapped.	GIS map and location-based viewing of biomass records.
Aggregation Coordination	The study will identify how hubs or cooperatives can consolidate records and monitor inventory.	Hub inventory, aggregation records, storage notes, quality notes, and inventory updates.
Marketplace Inquiry Support	The study will identify buyer information needs and inquiry limits.	Validated listings, buyer inquiry form, inquiry status, and communication records without payment or procurement.
Reporting and Planning Support	LGU and institutional users will identify needed reports and summaries.	Dashboards, reports, activity logs, transaction monitoring, and audit logs.

3.7 Procedure for Examining the Current Problem

The first research question will be answered by examining how biomass information is currently collected, stored, updated, validated, and shared among

stakeholders. This step is needed because AgriCycle should respond to an actual information problem, not only to assumed system features.

3.7.1 Document Review

The researchers will review the Butuan City Energy Development Plan, the biomass pre feasibility study, related literature, related systems, and available local records. The review will focus on biomass sources, residue types, current uses, availability concerns, data gaps, and stakeholder coordination needs. This will help identify the information that should be captured by the system.

3.7.2 Stakeholder Needs Assessment

The researchers will administer a needs assessment questionnaire to selected participants. The questions will cover the present way of recording biomass data, frequency of updates, information being recorded, availability checking, problems in sharing information, and reporting needs. The answers will be summarized to identify repeated issues and common needs.

3.7.3 Key Informant Interview

Key informant interviews will be conducted with selected participants who can explain the process in greater detail. The interview will clarify current use of biomass, uncertainty of availability, storage condition, quality concerns, validation practices, and buyer or institutional inquiries. Interview data will be grouped into themes using thematic analysis, which means identifying repeated meanings in responses and organizing them into clear themes (Braun & Clarke, 2006).

3.7.4 Expected Output

The output of this stage will be a problem analysis and a validated list of system needs. These outputs will guide the As-Is BPMN, Proposed BPMN, requirements matrix, user roles, data fields, Agile backlog, data entities, and system evaluation tasks.

3.8 Requirements Analysis and System Design

The second research question will be answered by converting the problem findings into system requirements and design documents. The system design will focus on the functions needed for biomass availability information management, aggregation records, GIS mapping, marketplace inquiry coordination, reports, and audit logs.

3.8.1 Requirements Analysis

The requirements analysis will translate stakeholder needs into specific system requirements. The researchers will prepare a requirements matrix that shows the problem, required data, system function, user role, and expected output. Data quality will be considered because users need information that is accurate, timely, complete, credible, and useful for decision-making (Wang & Strong, 1996; ISO/IEC, 2008).

Table 20. Sample Requirements Traceability Matrix

Problem Found	System Requirement	Main User	Expected Output
Biomass records are scattered or inconsistent.	Provide standard fields for biomass type, quantity, source, location, current use, availability period, storage status, quality note, and remarks.	Cooperative, rice mill, LGU, or authorized encoder.	Organized biomass records.
Availability is difficult to verify.	Provide validation status, validation notes, and user roles for checking records.	Validator or authorized administrator.	Checked records with clear status.
Source locations are hard to view together.	Provide GIS mapping for biomass sources, hubs, mills, and collection points.	LGU, cooperative, buyer, or institutional user.	Map view for planning and coordination.
Aggregation and storage updates are not consolidated.	Provide aggregation and inventory records with quality and storage notes.	Hub or cooperative user.	Updated inventory and aggregation summary.
Buyer inquiries are not tracked clearly.	Provide a marketplace inquiry function without payment, contract, or procurement processing.	Buyer or institutional user.	Recorded and traceable inquiries.

Reports are difficult to prepare manually.	Provide summary reports, dashboard records, and activity logs.	LGU, administrator, or institutional user.	Reports for monitoring and documentation.

3.8.2 Process and System Design Diagrams

The diagrams will follow a clear order: current process, proposed process, users and functions, database structure, and system structure. BPMN will be used to show the process flow because it is a standard notation for business process modeling (Object Management Group, 2014). UML will be used to show the interaction between users and the system (Object Management Group, 2017).

The study will prepare the following design documents: As-Is BPMN, Proposed BPMN, Use Case Diagram, Entity Relationship Diagram, System Architecture, data dictionary, and interface plan. These documents will guide the Agile backlog and development iterations.

3.8.3 Main Data Entities

To make the biomass availability information clear, the system will be designed around specific data entities. These entities will also guide the ERD, data dictionary, forms, validation rules, reports, and evaluation tasks.

Table 21. Main Data Entities of AgriCycle

Data Entity	Main Data Included	Purpose in the System
User	Name or user code, role, organization, contact details when needed, account status, and access level.	Controls who can encode, validate, view, inquire, or generate reports.
Biomass Source	Source name, source type, barangay or address, coordinates, owner or contributor group, and source description.	Identifies where biomass may come from.
Biomass Record	Biomass type, estimated quantity, unit, current use, availability period, storage condition, visible quality note, remarks, and record status.	Stores the main biomass availability information.

Validation Record	Biomass record checked, validator, validation date, status, notes, and action history.	Shows whether biomass information has been checked before use in listings or reports.
Location	Barangay, address description, latitude, longitude, source point, hub point, mill point, or collection point.	Supports GIS mapping and location-based review.
Aggregation Hub	Hub or cooperative name, linked biomass records, inventory updates, storage notes, quality notes, and available quantity.	Supports consolidation of biomass records and hub level monitoring.
Buyer Inquiry	Buyer or institutional user, listing viewed, purpose, quantity interest, message, date, and inquiry status.	Records marketplace coordination without payment, bidding, or legal contract processing.
Report and Audit Log	Generated summaries, user actions, status changes, listing changes, inquiry records, and timestamps.	Supports transparency, documentation, and planning review.

3.8.4 Core and Supporting System Requirements

The system requirements are grouped into core and supporting functions. Core functions are the minimum functions needed to prove the CB-BAIM Model. Supporting functions complete the coordination workflow but may be simplified if access, time, or technical limits arise during development.

Table 22. Core and Supporting System Requirements

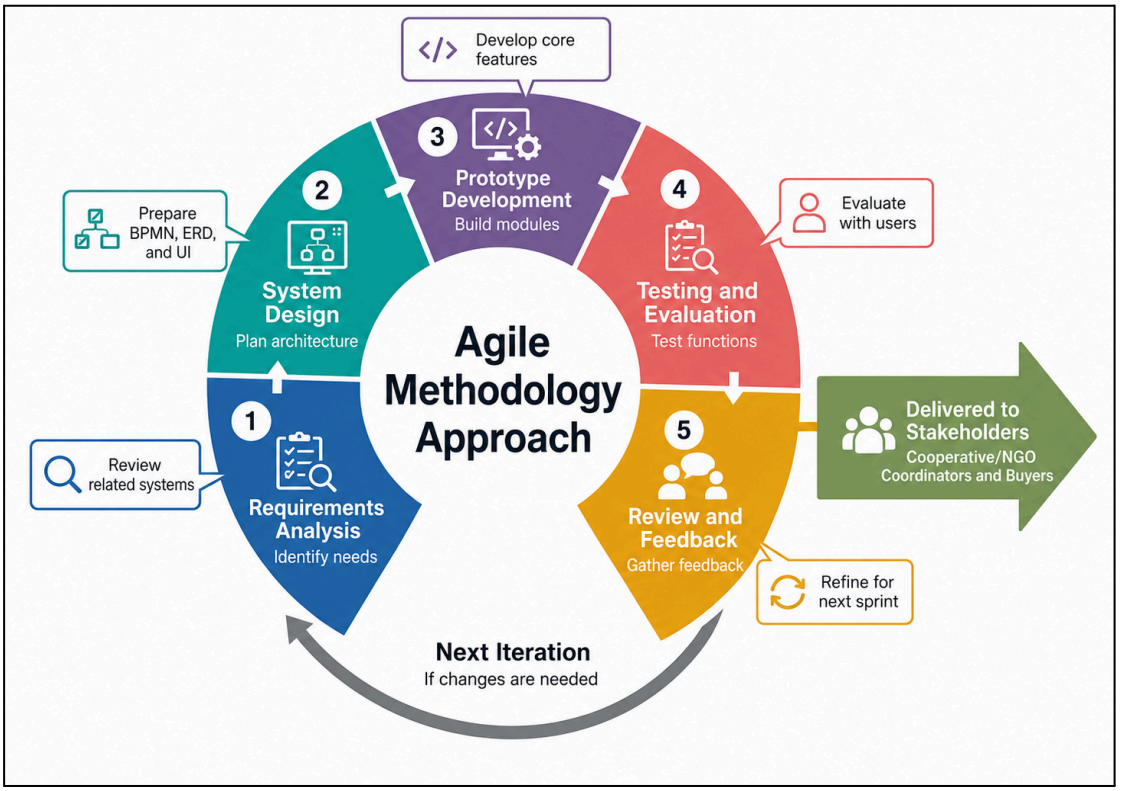
Priority	Function	Main Requirement	Purpose in the Study
Core	Biomass Data Collection and Management	Record biomass type, estimated quantity, source location, availability period, current use, contributor, storage status, quality note, and remarks.	Addresses scattered and inconsistent biomass records.
Core	Validation Workflow	Allow authorized users to mark records as available, unavailable, reserved, already used, unsuitable, or pending validation.	Prevents unverified data from being treated as usable supply.
Core	GIS Mapping and Visualization	Show biomass source locations, rice mills, cooperative hubs, and collection points.	Supports location based planning and coordination.
Core	Data Quality Checks	Use required fields, status rules, consistent units, and clear validation notes.	Supports completeness, accuracy, consistency, and credibility of information.
Supporting	Aggregation and	Record hub-level biomass inventory, storage status, quality notes, aggregation activity, and updates.	Supports cooperative based consolidation and practical availability tracking.

	Inventory Monitoring		
Supporting	Marketplace Listing and Buyer Inquiry	Allow validated listings to be viewed by authorized buyers and allow inquiries to be recorded and tracked.	Supports coordination without handling payment, bidding, procurement, or legal contracts.
Supporting	Reports, Dashboard, and Audit Logs	Generate summaries and record system actions, status changes, listings, inquiries, and validation activities.	Supports transparency, LGU planning, BEDP monitoring, and documentation.

3.9 Agile System Development Methodology

The study will use Agile methodology for the actual development of AgriCycle. Agile was selected because the system can be built in smaller parts, checked regularly, and improved before final evaluation. This is more suitable than developing the whole system at once because AgriCycle has several connected modules.

Figure 3. Agile Methodology Approach.



Design Science Research will still guide the overall research process, while Agile will guide the development work. In simple terms, DSR explains why the study

creates and evaluates a system artifact. Agile explains how the system will be built and improved through iterations.

The study will have one preparation phase and four Agile development iterations. The preparation phase is used to organize the backlog and system plan. It is not counted as a development iteration. The four Agile iterations are the actual development sprints. Each sprint will include sprint planning, design, coding, internal testing, feedback recording, and revision.

To control the scope, the first two sprints will focus on the core functions: user access, biomass records, validation, data quality checks, and GIS visibility. The third and fourth sprints will add supporting functions such as aggregation, inquiry, reports, and audit logs. This protects the study from claiming a full commercial marketplace or full logistics platform.

Table 23. Agile Development Iterations for AgriCycle

Phase or Sprint	Main Focus	Expected Output	Estimated Duration
Preparation Phase	Review the problem background, organize user needs, define data fields, identify user roles, prepare process flow, prepare data dictionary, and create the initial interface plan.	Product backlog, approved requirements list, data dictionary, user roles, and development plan.	3 weeks
Sprint 1	Develop login, role based access, biomass source profiles, biomass data encoding, standard fields, required field controls, and basic record management.	Working user access, biomass source records, and biomass data records.	2 weeks
Sprint 2	Develop validation status, validation notes, approval flow, data quality checks, map interface, source locations, rice mill locations, cooperative hubs, and collection points.	Working validation workflow, data quality controls, and GIS visualization functions.	2 weeks
Sprint 3	Develop hub inventory, aggregation records, storage status, quality notes, validated listings, buyer inquiry form, and inquiry status tracking.	Working aggregation, inventory, listing, and inquiry functions.	2 weeks
Sprint 4	Develop reports, dashboard summaries, transaction logs, activity logs, exportable records where applicable, interface review, and full system checking.	Integrated system ready for internal testing and user evaluation.	2 weeks

At the end of each sprint, the researchers will review the completed functions against the sprint goals and the requirements matrix. Issues will be recorded in the

sprint review notes. Revisions will be applied before moving to the next sprint when the issue affects the next module. Minor interface or wording revisions may be grouped for the final integration sprint. The researchers will use the Agile idea of short iterations, checking, feedback, and revision.

3.10 Procedure for System Testing and Evaluation

The third research question will be answered after the Agile development sprints are completed. The evaluation will focus on whether AgriCycle provides the needed functions, whether intended users can use it clearly, and whether the biomass information is complete, accurate, consistent, and credible enough for coordination. The formal software quality criteria will be Functional Suitability and Usability based on ISO/IEC 25010 (ISO/IEC, 2011).

3.10.1 Internal Testing

Before user evaluation, the researchers will conduct internal testing. The purpose is to check whether the main system functions work before the system is shown to respondents. The researchers will prepare test cases for common user actions such as adding a biomass record, validating a record, viewing the map, updating inventory, creating a listing, submitting an inquiry, generating reports, and reviewing audit logs.

Table 24. Sample Internal Test Cases

Test Case	Procedure	Expected Result
Add biomass record	The user enters biomass type, quantity, location, current use, availability period, storage status, and quality note.	The record is saved with complete required fields and pending validation status.
Validate biomass record	The authorized user reviews the record and updates the validation status.	Record status changes correctly and the action is logged.
View GIS map	The user opens the map and filters source locations, hubs, mills, or collection points.	Locations appear correctly and support coordination review.
Update aggregation record	The hub user updates inventory, storage status, quality note, or aggregation status.	Inventory record is updated and linked to the biomass source.

Create marketplace listing	An authorized user publishes a listing only from validated available biomass.	Listing is created and visible to authorized buyers.
Submit buyer inquiry	The buyer views a listing and sends an inquiry.	Inquiry is stored, linked to the listing, and visible to the assigned user.
Generate report	The user selects available report summaries.	The system produces a readable summary based on stored records.
Check audit log	Administrator reviews status updates, listings, inquiries, and validation actions.	The system shows traceable activity records.

3.10.2 User Evaluation

After internal testing, available qualified respondents will evaluate the system. The evaluation will begin with a short orientation about the purpose of the study, the user roles, the system flow, and the limits of the marketplace inquiry function. Respondents will then perform guided tasks or observe a guided demonstration depending on their role, access, and available time.

The task set will be adjusted to the participant role. For example, a cooperative user may focus on biomass records and aggregation, while a buyer may focus on validated listings and inquiry. Technical evaluators may review all modules through a guided walkthrough. After using or reviewing the system, they will answer the evaluation questionnaire and provide open-ended comments.

Table 25. Sample User Evaluation Tasks

Task	Participant Role Most Concerned	Task Description	Related Evaluation Area
Task 1	Encoder, cooperative user, rice mill user	Encode or review a biomass source and biomass record using required fields.	Functional Suitability and Usability.
Task 2	Validator, LGU, authorized administrator	Check a biomass record, add validation notes, and update the availability status.	Functional Suitability and Usability
Task 3	LGU, cooperative user, buyer, institutional user	View the GIS map and filter sources, hubs, mills, or collection points.	Functional Suitability and Usability

Task 4	Cooperative hub user	Update aggregation or inventory information, including storage and quality notes.	Functional Suitability and Usability
Task 5	Buyer or institutional user	View a validated listing and submitting an inquiry without payment or contract processing.	Functional Suitability and Usability
Task 6	LGU, administrator	Generate a report and review activity or audit log records.	Functional Suitability and Usability

3.10.3 Evaluation Criteria

Functional Suitability will determine whether the system provides the needed functions for biomass records, validation, GIS mapping, aggregation and inventory, buyer inquiry, reports, and audit logs. Usability will determine whether the system is understandable, easy to learn, easy to operate, and clear in terms of screens, labels, buttons, tables, maps, and reports. Data Quality will determine whether the stored biomass availability information is complete, accurate, consistent, and credible enough for coordination and planning.

Table 26. Evaluation Criteria for AgriCycle

Evaluation Area	Sub-Criterion	Application in AgriCycle
Functional Suitability	Functional completeness	The system provides the necessary functions for biomass records, validation, GIS mapping, inventory, marketplace inquiry, reports, and audit logs.
Functional Suitability	Functional correctness	The system produces correct outputs and status updates based on user actions.
Functional Suitability	Functional appropriateness	The system functions help users accomplish biomass information management and coordination tasks.
Usability	Appropriateness recognizability	Users can recognize the purpose of the system and its functions.
Usability	Learnability	Users can learn how to use the system with reasonable guidance.
Usability	Operability	Users can navigate, encode, update, view, inquire, and generate reports without unnecessary difficulty.
Usability	User interface clarity	Screens, labels, buttons, maps, tables, and reports are readable and understandable.

Usability	User error protection	The system reduces common input and status update errors through required fields, limits, and clear prompts.

3.11 Data Gathering Procedure

Data gathering will be conducted in phases. The process will begin with permission and preparation, followed by document review, stakeholder needs assessment, requirements analysis, Agile development, internal testing, user evaluation, data analysis, and final revision. This order connects the research problem, stakeholder data, system development, and evaluation results.

Table 27. Data Gathering and Development Procedure

Phase	Activity	Expected Output
Phase 1: Permission and Preparation	Secure approval from the adviser or institution and prepare letters, consent forms, and research instruments. Request access from concerned offices or organizations when needed.	Approved instruments and permission documents.
Phase 2: Document Review	Review the BEDP, biomass pre-feasibility study, related literature, related systems, and available local records.	Initial problem list, data fields, and system requirements.
Phase 3: Stakeholder Needs Assessment	Administer questionnaires and conduct interviews about biomass data practices, current uses, validation, storage, quality, reporting, and coordination.	Problem analysis, stakeholder needs, and user stories.
Phase 4: Requirements Analysis and System Design	Prepare requirements matrix, As-Is BPMN, Proposed BPMN, Use Case Diagram, ERD, System Architecture, interface plan, user roles, data dictionary, and reporting structure.	System design documents and Agile product backlog.
Phase 5: Agile Development	Build the system through four development sprints, with checking, feedback, and revision after each sprint.	Working system functions and sprint review notes.
Phase 6: Internal Testing	Check input errors, navigation problems, validation status, mapping display, inquiry flow, report generation, data quality rules, and audit logs.	Tested system ready for user evaluation.
Phase 7: User Evaluation	Allow available qualified users and evaluators to test or review the system and answer the evaluation questionnaire.	Functional Suitability and Usability.

Phase 8: Analysis and Final Revision	Compute results, analyze comments, identify issues, and apply necessary revisions.	Final system, evaluation results, conclusions, and recommendations.

3.12 Treatment and Analysis of Data

The data will be analyzed according to the three objectives. For the problem assessment, answers from the needs assessment and interviews will be summarized using frequency counts, descriptive interpretation, and thematic analysis. Common themes may include inconsistent records, delayed updating, unclear current use, lack of validation, storage concerns, location concerns, and weak buyer coordination. Thematic analysis will follow the process of identifying repeated meanings in the responses and grouping them into clear themes (Braun & Clarke, 2006).

For system design and development, the gathered data will be converted into system requirements, user roles, process flows, data fields, data entities, Agile backlog items, and system functions. The development outputs will be documented through sprint review notes, revision logs, and completed feature checklists. This will show how the system functions and design diagrams came from research findings and stakeholder needs.

For system evaluation, the results will be treated using weighted mean and standard deviation. The weighted mean will determine the average rating for each criterion. The formula will be $WM = \Sigma fx / N$, where WM refers to weighted mean, f refers to the frequency of responses, x refers to the rating value, and N refers to the total number of responses. The standard deviation will show how close or varied the responses are from the mean rating. Open ended comments will be grouped into usability issues, functional issues, data quality issues, content issues, and improvement suggestions.

Table 28. Interpretation Scale for System Evaluation

Range	Interpretation	Meaning
4.21 - 5.00	Strongly Agree / Highly Suitable	The system strongly meets the criterion.
3.41 - 4.20	Agree / Suitable	The system meets the criterion with minor improvement needed.
2.61 - 3.40	Moderately Agree / Moderately Suitable	The system partly meets the criterion but needs improvement.

1.81 - 2.60	Disagree / Less Suitable	The system has noticeable weakness in the criterion.
1.00 - 1.80	Strongly Disagree / Not Suitable	The system does not meet the criterion.

3.13 Ethical Considerations and Data Privacy

The study will follow ethical research practices. Respondents will be informed about the purpose of the study, the voluntary nature of participation, the type of questions to be asked, and the expected use of their responses. Consent will be obtained before interviews, questionnaires, or system evaluation. Participants may refuse to answer questions or withdraw from the study without penalty.

Personal information will be limited to what is necessary for respondent grouping and system testing. Names of respondents will not be disclosed in the final report unless formal permission is given. Evaluation results will be reported in summary form. Access to system records during testing will be limited to authorized users. These practices are consistent with the general principles of privacy, consent, and responsible data handling under the Data Privacy Act of 2012 in the Philippines (Republic Act No. 10173, 2012).

The study will also avoid misrepresenting biomass availability. A biomass listing in the system does not mean that a legal sale, payment, procurement, or contract has already happened. The marketplace feature is only for inquiry and coordination. Final agreements, price negotiation, procurement, transport arrangements, and payments will remain outside the system and must be handled by the involved parties according to their own rules and applicable policies.

3.14 Summary of Methodology

In summary, the study will use a descriptive-developmental research design guided by Design Science Research and developed through Agile methodology. It will begin with document review and stakeholder needs assessment, then proceed to requirements analysis, system design, Agile development sprints, internal testing, user evaluation, data analysis, and final revision.

The methodology operationalizes the CB-BAIM Model by examining the

biomass information problem, translating the model into system functions, and evaluating whether the system is functionally suitable, usable, and supported by quality biomass availability information. The final output will be AgriCycle and its documented evaluation results for biomass availability information management, aggregation, and marketplace inquiry coordination.

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APPENDICES
DATA-GATHERING INSTRUMENTS FOR AGRICYCLE BUTUAN

Study Title: AgriCycle: A Community-Based Biomass Information Management and Market Linkage System

These instruments are prepared to gather data needed for the Biomass Availability Information Framework of AgriCycle. The framework does not treat theoretical biomass potential as automatically usable supply. It checks whether biomass is actually available by considering existing uses, source location, seasonality, storage condition, quality, transport concerns, and cooperative validation. The instruments remain focused on information management, aggregation, and marketplace coordination. They do not measure power plant feasibility, actual energy output, financial viability, or long-term environmental impact.

The questions are based on the study objectives, the Butuan City Energy Development Plan 2023-2050, the biomass pre-feasibility findings for Butuan City, and software quality evaluation standards. The BEDP identifies biomass as one of the city's renewable energy resources and supports energy data management and monitoring. The pre-feasibility study shows why practical availability must be checked because some residues are already used, scattered, costly to collect, or not enough for a stand-alone biomass facility (City Government of Butuan, 2023; Preferred Energy, Inc., 2025).

INSTRUMENT ALIGNMENT MATRIX

This matrix shows how each research question is answered by the data-gathering tools. The added direction is the Biomass Availability Information Framework, which converts scattered biomass records into checked and usable availability information.

Research Question	Objective	Needed Data / Output
RQ1: What are the core problems in biomass availability data collection, information management, and	Objective 1: Examine current procedures, record standardization, information sharing, reporting delays, existing uses, practical availability,	Stakeholder responses on current biomass data practices, availability concerns, and needed information fields.

coordination in Butuan City?	location, storage, quality, validation, and decision use.	
RQ2: How may AgriCycle be designed and developed to support biomass availability information management?	Objective 2: Design and develop system modules for standardized encoding, cooperative validation, GIS mapping, inventory and quality records, aggregation monitoring, marketplace coordination, reports, and audit logs.	System design and development outputs based on the Biomass Availability Information Framework.
RQ3: How does the developed AgriCycle system perform in terms of Functional Suitability and Usability?	Objective 3: Evaluate whether the prototype functions match the intended needs and whether users can understand and use the system.	User evaluation ratings and feedback after prototype walkthrough.

BIOMASS AVAILABILITY INFORMATION FRAMEWORK USED IN THE INSTRUMENT

Framework Stage	Use in Data Gathering and System Design
1. Data Capture	Record biomass type, estimated quantity, source barangay, contributor type, availability period, existing use, storage condition, quality observation, and location.
2. Practical Availability Check	Check whether biomass is usable for aggregation, already used internally, reserved, unavailable, spoiled, too wet, too far, or still uncertain.
3. Cooperative Validation	Allow authorized cooperative or hub personnel to confirm, reject, update, or mark biomass records for follow-up.
4. Information Integration	Organize validated records into inventory, GIS map, aggregation records, and marketplace listings.
5. Coordination Use	Use the records for buyer inquiries, cooperative reports, LGU monitoring, and waste-to-value planning.

APPENDIX A
INFORMED CONSENT FORM

This form follows the ethical requirement that participants must be informed before joining the study and that their participation must be voluntary. It also follows the Data Privacy Act of 2012 because the study may collect names, roles, contact details, and work-related responses (Philippine Health Research Ethics Board, 2022; Republic Act No. 10173, 2012). *(Kini nga form nagsunod sa ethical requirement nga kinahanglan ma-inform ang participants before sila moapil sa study, ug boluntaryo dapat ang ilang participation. Nagsunod usab kini sa Data Privacy Act of 2012 kay ang study mahimong mangolekta og ngalan, roles, detalye sa pag, ug responses nga related sa ilang work.).*

Purpose of Participation: You are invited to participate in the data-gathering, workflow validation, system walkthrough, or system evaluation for AgriCycle. Your answers will help the researchers understand how biomass information is recorded, checked, organized, mapped, and shared for aggregation and marketplace coordination. *(Katuyoan sa Pag-apil: Gi-invite kamo nga moapil sa data gathering, workflow validation, system walkthrough, o system evaluation para sa AgriCycle. Ang inyong answers makatabang sa researchers nga masabtan kung giunsa pag-record, pag-check, pag-organize, pag-map, ug pag-share ang biomass information* para sa *aggregation ug marketplace coordination.).*

Activities and Estimated Time: Participation may involve answering a questionnaire, joining a short workflow validation interview, reviewing a prototype walkthrough, and/or answering a system evaluation form. Each activity may take about 10 to 30 minutes, depending on the respondent role and the instrument used. *(Mga Buhaton ug Gibanabanang Oras: Ang inyong participation mahimong maglakip sa pagtubag og questionnaire, mubo nga workflow validation interview, pagtan-aw sa prototype walkthrough, ug/o pagtubag sa system evaluation form. Ang matag activity mahimong moabot og mga 10 hangtod 30 minutos, depende sa inyong role ug sa instrument nga gamiton.).*

Voluntary Participation and Right to Withdraw: Your participation is voluntary. You may refuse to answer any question or withdraw from the activity at

any time without penalty. You may also ask that your response be excluded before the data are summarized or anonymized. *(Boluntaryong Pag-apil ug Katungod sa Pag-undang: Boluntaryo ang inyong participation. Pwede kamo dili motubag sa bisan unsang question o mohunong bisan unsang oras nga walay penalty. Pwede usab ninyo hangyoon nga dili iapil ang inyong response basta wala pa ma-summary o ma-anonymize ang data.)*.

Possible Risks or Discomforts: The study involves minimal risk. The possible discomfort is limited to the time needed to answer the questions or the possibility that some questions may be unfamiliar. You may skip any question that you do not want to answer. *(Posibleng Risks o Discomforts: Gamay ra ang risk niini nga study. Ang possible discomfort mahimong tungod sa oras sa pagtubag o kung naay questions nga dili pamilyar. Pwede ninyo laktawan ang bisan unsang question nga dili ninyo gustong tubagon.)*.

Benefits: There may be no direct monetary benefit from participation. However, your responses may help improve the design of a biomass availability information system for cooperatives, LGU offices, buyers, and other authorized stakeholders in Butuan City. *(Benepisyo: Posibleng walay direct monetary benefit gikan sa inyong participation. Pero ang inyong responses makatabang sa pag-improve sa design sa biomass availability information system para sa cooperatives, LGU offices, buyers, ug uban pang authorized stakeholders sa Butuan City.)*.

Confidentiality and Data Privacy: Your identity will be kept confidential. Responses will be summarized for academic purposes only and will not be reported using your name unless permission is given. Questionnaires will use respondent codes whenever possible. Names and contact details will be used only for consent documentation, follow-up clarification, or ethics verification when necessary. *(Confidentiality ug Data Privacy: Protektahan ang inyong identity ug dili kini ipadayag. Ang responses i-summary para sa academic purposes lamang ug dili gamiton ang inyong name sa report gawas kung mosugot kamo. Respondent codes ang gamiton kung mahimo. Ang names ug contact details gamiton lamang para sa consent documentation, follow-up clarification, o ethics verification kung kinahanglan.)*.

Data Storage and Retention: Research data will be stored in password-protected digital files and/or secured physical folders accessible only to the researchers, adviser, and authorized academic reviewers. The data will be kept only for the period required for thesis completion and school research documentation, after which identifiable files will be deleted or securely disposed of. *(Data Storage ug Retention: Ang research data ibutang sa password-protected digital files ug/o luwas nga physical folders. Ang researchers, adviser, ug authorized academic reviewers ra ang maka-access niini. Tipigan lang ang data hangtod mahuman ang thesis ug ang kinahanglan nga school research documents. Human ana, ang files nga adunay impormasyon nga makaila sa participant i-delete o i-dispose sa luwas nga paagi.).*

Use of Audio Recording: No audio or video recording will be conducted unless the participant gives separate permission. If recording is not permitted, the researchers will rely only on written notes. *(Paggamit og Audio o Video Recording: Walay audio o video recording nga himuon gawas kung mohatag og separate permission ang participant. Kung dili mosugot sa recording, written notes ra ang gamiton sa researchers.).*

Limitations of the Study: The system evaluation will not involve online payment, legal contract processing, public bidding, route optimization, GPS tracking, or actual commercial transactions. *(Limitasyon sa Study: Ang system evaluation dili maglakip sa online payment, legal contract processing, public bidding, route optimization, GPS tracking, o actual commercial transactions.).*

Researcher Contact: For questions about the study, participants may contact:

Researchers Names: Nitron Ford M. Beatingo, Aiah Lei Danielle M. Entrampas, and Arvin S. Jimenez;

Contact Number/Email: 09751121029/arvin.jimenez@g.msuiit.edu.ph;
09261496570/aiahleidanielle.entrampas@g.msuiit.edu.ph;
09551025364/nitronford.beatingo@g.msuii.edu.oh.

Thesis Adviser: Mrs. Mary Ann Gliefen Bermudo-Andrade;

Department/School: College of Computer Studies (CCS) at Mindanao State University-Iligan Institute of Technology.

Consent Statement: I have read and understood the information above. I voluntarily agree to participate in this study and allow my responses to be used for academic research purposes under the confidentiality conditions stated in this form. *(Consent Statement: Nabasa ug nasabtan nako ang impormasyon sa taas. Kusa ra akong miuyon nga moapil niini nga study, ug mosugot ko nga gamiton ang akong mga tubag para sa academic research. Nasabtan usab nako nga protektahan ug dili ipadayag ang akong personal nga impormasyon sumala sa nakasulat niini nga form.)*.

Item	**Response**
Participant Name / Code Name	
Role / Organization Type	
Contact Number / Email (optional)	
Date	

APPENDIX B
RESPONDENT PROFILE FORM

This form is used to identify respondents who can give useful information about biomass data collection, cooperative work, rice milling, aggregation, buying, LGU planning, or system use. Purposive selection is appropriate because the study needs respondents who have direct knowledge of the subject (Patton, 2015). Digital agriculture studies also need to consider differences in user roles, access, and digital familiarity (Trendov et al., 2019). Only adult respondents who are 18 years old and above will be invited to participate. *(Kini nga form gamiton para mahibal-an kung kinsa nga respondents ang makahatag og importanteng impormasyon bahin sa biomass, trabaho sa cooperative, rice milling, pagtigom sa biomass, pagpamalit, pagplano sa LGU, o paggamit sa system. Pilion sa study ang mga respondents nga adunay kahibalo o experience bahin niini. Mga 18 years old pataas ra ang pwede moapil.).*

Item	**Response**
Respondent Code / Name	
Respondent Group	[ ] Cooperative Hub Manager / Cooperative Personnel [ ] Rice Mill Personnel [ ] Aggregation Center Operator [ ] Farmer Representative [ ] NGO Coordinator [ ] LGU / Government Representative [ ] Institutional Buyer [ ] Renewable Energy / Biomass User [ ] Agricultural Enterprise [ ] Researcher / Academic Stakeholder [ ] System Administrator / Technical Staff [ ] Other: _____
Barangay / Area of Work	
Years of Relevant Experience	[ ] Less than 1 year [ ] 1-3 years [ ] 4-6 years [ ] More than 6 years

Familiarity with Agricultural Biomass	[ ] Very Low [ ] Low [ ] Moderate [ ] High [ ] Very High
Familiarity with Digital Systems	[ ] Very Low [ ] Low [ ] Moderate [ ] High [ ] Very High
Involvement in Biomass Data Collection	[ ] Yes [ ] No [ ] Occasionally
Involvement in Biomass Buying / Utilization	[ ] Yes [ ] No [ ] Occasionally
Role in This Study Activity	[ ] Problem Assessment Questionnaire [ ] System Walkthrough [ ] System Evaluation
Date of Participation	

APPENDIX C
BIOMASS AVAILABILITY AND INFORMATION MANAGEMENT QUESTIONNAIRE

This questionnaire asks about how biomass information is recorded, shared, delayed, validated, and used. It also asks about practical availability because the Butuan biomass pre-feasibility study showed that theoretical biomass potential does not automatically mean usable supply. Rice husks, coconut husks, coconut shells, and wood residues may already have existing uses, while rice straw and other residues may be difficult to collect because of timing, distance, labor, storage, and quality concerns (City Government of Butuan, 2023; Preferred Energy, Inc., 2025). *(Kini nga questionnaire nangutana kung giunsa pag-record, pag-share, pag-update, pag-check, ug paggamit ang impormasyon bahin sa biomass. Nangutana usab kini kung naa ug magamit pa ba gyud ang biomass kay dili pasabot nga kung naa kini sa estimate, pwede na dayon gamiton. Ang uban mahimong gigamit na, lisod kolektahon, layo ang source, naa ra depende sa season, kulang ang trabahante, o adunay problema sa pagtipig ug quality.).*

The five-point scale is based on Likert-type response measurement. The questions also use data quality concerns such as completeness, consistency, currentness, and credibility, which are important in information systems (International Organization for Standardization, 2008; Likert, 1932). *(Ang five-point scale naggamit og Likert-type nga pagtubag. Gina-check usab sa mga pangutana kung kompleto ba ang data, parehas ba ang pag-record, updated ba, ug kasaligan ba ang impormasyon.).*

Instruction: Please rate each statement based on your experience or knowledge of biomass data, agricultural residues, cooperative work, LGU planning, buyer inquiry, or system use. If a statement does not apply to your role, you may choose Neutral or leave a note for the researcher. *(Instruction: Palihog i-rate ang matag pahayag base sa inyong experience o nahibaloan bahin sa biomass data, agricultural residues, trabaho sa cooperative, LGU planning, buyer inquiries, o paggamit sa system. Kung dili kini applicable sa inyong role, pwede pilion ang Neutral o magbutang og mubo nga note para sa researcher.).*

Rating	Interpretation
5	Strongly Agree *(Uyon Kaayo)*
4	Agree *(Uyon)*
3	Neutral
2	Disagree *(Dili Uyon)*
1	Strongly Disagree *(Dili Gyud Uyon)*

A. Existing Data Collection Procedures for Biomass Supply

(Paagi Karon sa Pagkolekta ug Pag-record sa Impormasyon Bahin sa Biomass).

It checks whether biomass information is already recorded in a way that can support aggregation and coordination. *(Tan-awon niini kung sakto ba ang pag-record sa biomass information aron mas sayon ang pagtigom sa biomass ug pag-coordinate sa mga involved nga tawo o grupo.).*

No.	Statement	5	4	3	2	1
1	Biomass availability is currently recorded in an organized manner. *(Sa pagkakaron, tarong ug organisado ang pag-record kung unsang biomass ang naa ug makuha.).*					
2	Our organization has a clear procedure for collecting biomass-related information. *(Ang among organisasyon adunay klarong paagi sa pagkolekta og impormasyon bahin sa biomass.).*					
3	Information on biomass type, estimated volume, source location, and availability period is regularly recorded. *(Kanunay gina-record ang klase sa biomass, estimated amount, asa kini gikan, ug kanus-a kini available.).*					
4	Current biomass records are easy to access when needed. *(Sayon ra pangitaon ang mga rekord sa biomass kung kinahanglan.).*					

5	Existing data collection procedures are sufficient for monitoring biomass supply. *(Igo ang among paagi karon sa pagkolekta og data para ma-monitor ang supply sa biomass.).*					

B. Consistency and Standardization of Biomass Availability Records

It checks whether records use the same fields, units, and descriptions across sources and hubs. *(Tan-awon niini kung parehas ba ang mga detalye, units, ug descriptions nga gigamit sa biomass records sa lain-laing sources ug hubs.).*

No.	Statement	5	4	3	2	1
6	The same data fields are used when recording biomass information across collection points. *(Parehas nga mga detalye ang gina-record bisan asa nga collection point.).*					
7	Biomass records use standard categories for biomass type. (*Parehas nga categories ang gigamit sa pag-classify sa klase sa biomass.).*					
8	Biomass quantity or volume is recorded using consistent units of measurement. *(Parehas nga unit, sama sa kilo o tonelada, ang gamiton sa pag-record sa kadaghanon sa biomass.).*					
9	Biomass quality observations are recorded using consistent descriptions. *(Parehas ang paagi sa pag-describe sa quality sa biomass.).*					
10	Availability status is recorded consistently across different sources or hubs. *(Parehas ang paagi sa pag-record kung available pa ba ang biomass sa lain-laing source o hub.).*					

C. Fragmentation in Biomass Information Sharing

(Pagkatag-katag sa Biomass Information).

It checks whether biomass information is scattered among farmers, rice mills, cooperatives, buyers, LGU offices, and other stakeholders. *(Tan-awon niini kung nagkatag ba ang biomass information sa mga farmers, rice mills, cooperatives, buyers, LGU offices, ug uban pang involved nga tawo o grupo.).*

No.	Statement	5	4	3	2	1
11	Biomass information is currently scattered across different people, offices, records, or communication channels. *(Sa pagkakaron, nagkatag ang biomass information sa lain-laing tawo, opisina, records, o paagi sa pag-share sa information.).*					
12	Farmers, cooperatives, rice mills, buyers, and institutional stakeholders do not always share the same biomass information. *(Ang farmers, cooperatives, rice mills, buyers, ug uban pang involved nga organizations dili pirmi parehas og biomass information.).*					
13	Buyers have difficulty finding updated information about available biomass supply. *(Lisod usahay para sa mga buyer nga makakita og updated information bahin sa biomass nga naa ug makuha.).*					
14	Cooperatives or hubs have difficulty consolidating biomass information from different barangays or collection points. *(Lisod para sa cooperatives o hubs nga tigumon ang biomass information gikan sa lain-laing barangay o collection points.).*					
15	A centralized system is needed to organize and share biomass-related information among stakeholders. *(Kinahanglan og usa ka centralized system para ma-organize ug ma-share ang biomass information sa mga involved nga tawo o grupo.).*					

D. Delays and Inefficiencies in Biomass Data Capture and Reporting

(Pagkalangan ug Kalisod sa Pag-record ug Pag-report sa Biomass Data)

It checks whether late information affects collection, aggregation, and buyer inquiries. *(Tan-awon niini kung ang nalangan nga impormasyon makaapekto sa pagkolekta, pagtigom sa biomass, ug buyer inquiries.)*.

No.	Statement	5	4	3	2	1
16	Biomass availability records are updated on time. *(Ang mga rekord kung naa pa ang biomass gina-update sa saktong oras.)*.					
17	Delays occur in reporting biomass quantity, location, or availability. *(Usahay malangan ang pag-report sa kadaghanon, lugar, o pagkanaa sa biomass.)*.					
18	Delayed biomass information affects collection or aggregation planning. *(Ang nalangan nga impormasyon sa biomass makaapekto sa planning sa pagkolekta o pagtigom niini.)*.					
19	Delayed reporting affects buyer inquiries or sourcing decisions. *(Ang nalangan nga pag-report makaapekto sa mga pangutana ug desisyon sa buyer kung asa mokuha og biomass.)*.					
20	A digital system can help improve the timeliness of biomass data capture and reporting. *(Ang digital nga system makatabang nga mas paspas ug on time ang pag-record ug pag-report sa datos sa biomass.)*.					

E. Existing Uses and Practical Biomass Availability

(Kasamtangang Paggamit ug Practical Availability sa Biomass)

This section supports the Biomass Availability Information Framework. It checks whether biomass is still available after considering existing uses. This is needed because the pre-feasibility study found that rice husks may already be used by

rice millers for heating, drying, and domestic cooking, while other residues may also have current local uses (Preferred Energy, Inc., 2025). *(Kini nga section makatabang sa Biomass Availability Information Framework. Tan-awon niini kung available pa ba ang biomass human ma-check kung gigamit na ba kini sa lain nga paagi. Importante kini kay nakita sa pre-feasibility study nga ang rice husks usahay gigamit na sa rice millers para sa pagpainit, pagpauga, ug pagluto sa balay. Ang ubang biomass residues mahimo usab nga gigamit na sa lokal nga lugar (Preferred Energy, Inc., 2025).*

No.	Statement	5	4	3	2	1
21	Some biomass residues are already used by farmers, rice millers, or local users. *(Naay mga biomass residue nga gigamit na sa farmers, rice millers, o mga tawo sa lugar.).*					
22	Rice husks are often retained or used by rice millers for drying, heating, or domestic cooking. *(Ang ipa sa humay kasagarang gitipigan o gigamit sa rice millers para sa pagpauga, pagpainit, o pagluto sa balay.).*					
23	Not all recorded biomass residues are actually available for collection or selling. *(Dili tanan biomass residue nga naa sa record available pa para kolektahon o ibaligya.).*					
24	Current records do not clearly show whether biomass is available, already used, reserved, or unavailable. *(Ang current records dili pirmi klaro kung available pa ba ang biomass, gigamit na, gitagana na para sa lain, o dili na makuha.).*					
25	A system is needed to separate theoretical biomass estimates from practical biomass availability. *(Kinahanglan og system nga makapakita sa kalainan sa estimated biomass ug sa biomass nga naa ug pwede pa gyud kolektahon o gamiton.).*					

F. Seasonality, Collection Timing, and Availability Period

(Panahon sa Pagka-available, Pagkolekta, ug Availability Period sa Biomass)

This section checks when biomass becomes available. Timing matters because rice straw, coconut residues, corn residues, and other agricultural wastes may appear at different periods. *(Kini nga section tan-awon kung kanus-a available ang biomass. Importante ang timing kay ang rice straw, coconut residues, corn residues, ug uban pang agricultural waste mahimong available sa lain-laing panahon.).*

No.	Statement	5	4	3	2	1
26	Biomass availability changes depending on crop season or harvest schedule. *(Mausab-usab kung naa ang biomass depende sa panahon sa tanom o iskedyul sa ani.).*					
27	Collection timing affects whether biomass can still be used for aggregation. *(Ang oras o panahon sa pagkolekta makaapekto kung matigom ug magamit pa ba ang biomass.).*					
28	Rice straw availability is difficult to monitor because it depends on harvesting activities. *(Lisod ma-monitor kung naa pa ang uhot sa humay kay nagdepende kini sa panahon sa pag-ani.).*					
29	Biomass records should include the expected availability period or collection window. *(Ang mga rekord sa biomass kinahanglan magpakita kung kanus-a kini expected nga naa ug unsang panahona kini pwede kolektahon.).*					
30	A system that records availability periods can help improve aggregation planning. *(Ang system nga nag-record kung kanus-a naa ang biomass makatabang sa mas maayong planning sa pagtigom niini.).*					

G. Transportation, Location, and Collection Concerns

(Transport, Lokasyon, ug mga Problema sa Pagkolekta)

This section checks whether location and transport affect practical availability. The pre-feasibility study noted that some residues are too spread out for feasible collection (Preferred Energy, Inc., 2025). *(Kini nga section tan-awon kung ang lokasyon ug transport makaapekto ba kung makolekta pa gyud ang biomass. Nakita sa pre-feasibility study nga ang ubang residues nagkatag sa lain-laing lugar, mao nga lisod kini kolektahon (Preferred Energy, Inc., 2025).*

No.	Statement	5	4	3	2	1
31	Distance from farms, mills, or collection points affects biomass collection. *(Ang kalayo gikan sa farms, mills, o collection points makaapekto sa pagkolekta sa biomass.).*					
32	Lack of transport makes biomass aggregation difficult. *(Kung kulang o walay transport, mas lisod ang pagtigom sa biomass.).*					
33	Road access or hauling concerns affect whether biomass can be collected. *(Ang kahimtang sa dalan ug problema sa paghakot makaapekto kung makolekta ba ang biomass.).*					
34	Biomass records should show source location and nearest possible aggregation point. *(Ang biomass records kinahanglan magpakita kung asa gikan ang biomass ug asa ang pinakaduol nga lugar nga pwede kini tigumon.).*					
35	GIS mapping can help identify source areas, cooperative hubs, and collection points. *(Ang GIS mapping makatabang sa pagpakita kung asa ang source areas, cooperative hubs, ug collection points.).*					

H. Storage, Moisture, and Quality Concerns

This section checks quality-related information. Biomass may exist but may not be useful if it is wet, spoiled, mixed with contaminants, or not properly stored. *(Kini*

nga section tan-awon ang quality sa biomass. Bisan naa pa ang biomass, basin dili na kini magamit kung basa, nadaot, nasagolan og hugaw o laing materials, o dili sakto ang pagtipig.).

No.	Statement	5	4	3	2	1
36	Storage limitations affect whether biomass can be kept for future use. *(Ang kakulang sa storage makaapekto kung matipigan pa ba ang biomass para magamit sa umaabot.).*					
37	Moisture conditions affect the usefulness of biomass residues. *(Ang pagkabasa sa biomass residue makaapekto kung magamit pa ba kini.).*					
38	Quality observations are needed before biomass is listed for aggregation or buyer inquiry. *(Kinahanglan i-check una ang quality sa biomass bago kini ibutang sa listahan para tigumon o ipakita sa buyers.).*					
39	Biomass records should include storage condition and quality notes. *(Ang biomass records kinahanglan adunay impormasyon bahin sa storage condition ug quality sa biomass.).*					
40	A system should allow users to update quality observations when biomass condition changes. *(Ang system kinahanglan makapahimo sa users nga mo-update sa quality information kung mausab ang condition sa biomass.).*					

I. Cooperative-Based Validation Workflow

(Pag-check ug Pag-validate sa Biomass Data pinaagi sa Cooperative)

This section checks the proposed validation process. The study uses a cooperative-based model because smallholder farmers may not always directly use the system, while authorized hubs can encode and check records. *(Kini nga section tan-awon ang proposed nga proseso sa pag-check ug pag-validate sa biomass data.*

Gigamit sa study ang cooperative-based model kay dili tanan smallholder farmers pirmi makagamit diretso sa system. Ang authorized hub personnel maoy pwede mo-encode ug mo-check sa records.).

No.	Statement	5	4	3	2	1
41	Biomass information should be checked by authorized cooperative or hub personnel before it is used. *(Kinahanglan i-check una ang biomass information sa authorized cooperative o hub personnel bago kini gamiton.).*					
42	The system should show whether a biomass record is pending, verified, unavailable, reserved, or available. *(Ang system kinahanglan magpakita kung ang biomass record wala pa na-check, verified na, dili available, gitagana na para sa lain, o available.).*					
43	Cooperative hub managers should be able to validate biomass data from farmers or collection points. *(Ang cooperative hub managers kinahanglan makacheck ug maka-validate sa biomass data gikan sa farmers o collection points.).*					
44	Validated biomass records are more useful for buyer inquiries and LGU reporting. *(Mas useful ang na-check ug na-validate nga biomass records para sa buyer inquiries ug LGU reporting.).*					
45	A validation workflow can help reduce inaccurate or outdated biomass listings. *(Ang klarong proseso sa pag-check ug pag-validate makatabang nga maminusan ang sayop o outdated nga biomass listings.).*					

J. Data Use for LGU, BEDP Monitoring, and Waste-to-Value Planning

(Paggamit sa Data para sa LGU, BEDP Monitoring, ug Waste-to-Value Planning)

This section checks how the data may be used by the LGU and other institutions. The BEDP states the need for energy data infrastructure, monitoring, updated resource evaluations, and renewable energy planning (City Government of

Butuan, 2023). *(Kini nga section tan-awon kung unsaon paggamit sa data sa LGU ug uban pang institutions. Sumala sa BEDP, importante nga adunay sakto ug updated nga energy data para sa monitoring, pag-check sa available resources, ug planning para sa renewable energy (City Government of Butuan, 2023).*

No.	Statement	5	4	3	2	1
46	Biomass data can help the LGU update local biomass resource information. *(Ang biomass data makatabang sa LGU nga ma-update ang information bahin sa biomass resources sa lugar.).*					
47	Biomass availability records can support local waste-to-value planning. *(Ang records kung unsang biomass ang available makatabang sa waste-to-value planning sa lugar.).*					
48	GIS-based biomass records can help LGU offices identify source areas and possible aggregation points. *(Ang GIS-based biomass records makatabang sa LGU nga makita kung asa gikan ang biomass ug asa kini pwede tigumon.).*					
49	Verified biomass data can support monitoring of agricultural residues in Butuan City. *(Ang na-check ug verified nga biomass data makatabang sa pag-monitor sa agricultural residues sa Butuan City.).*					
50	The proposed system can provide useful reports for LGU planning, cooperatives, and institutional stakeholders. *(Ang proposed system makahatag og useful reports para sa LGU planning, cooperatives, ug uban pang involved nga institutions.).*					

Open-Ended Questions

These questions allow respondents to explain problems, local practices, and needed system fields that may not be fully captured by the rating items. Responses may be grouped by theme following Braun and Clarke (2006). *(Kini nga mga*

pangutana naghatag og chance sa respondents nga ma-share ang mga problema nga ilang na-experience, kasagarang paagi nga ilang ginabuhat sa ilang lugar, ug uban pang impormasyon nga basin wala naapil sa rating questions. Ang ilang mga tubag mahimong ipundok base sa parehas nga tema sumala kang Braun ug Clarke (2006).

No.	Question	Response
1	How do you currently collect or record biomass-related information? *(Giunsa ninyo karon pagkolekta o pag-record sa impormasyon bahin sa biomass?).*	
2	What types of biomass information do you usually record? *(Unsang klase sa impormasyon bahin sa biomass ang kasagaran ninyong gina-record?).*	
3	What are the main limitations of your current data collection process? *(Unsa ang kasagarang problema o kakulangan sa inyong paagi karon sa pagkolekta og data?).*	
4	What data fields should be standardized in the proposed system? *(Unsang mga detalye sa data ang kinahanglan parehas ang format o paagi sa pag-record sa proposed system?).*	
5	What problems occur when biomass records are incomplete or inconsistent? *(Unsang mga problema ang mahitabo kung kulang o dili parehas ang biomass records?).*	
6	How is biomass information currently shared among farmers, cooperatives, rice mills, buyers, LGUs, and other stakeholders? *(Giunsa karon pag-share ang biomass information sa farmers, cooperatives, rice mills, buyers, LGUs, ug uban pang involved nga grupo?).*	
7	What makes biomass information difficult to consolidate?	

	(Unsa ang nakapalisod sa pagtigom ug paghiusa sa biomass information gikan sa lain-laing sources?).	
8	What usually causes delays in recording or reporting biomass availability? *(Unsa kasagaran ang hinungdan nganong madugay ang pag-record o pag-report kung available ba ang biomass?).*	
9	What information do you need before making decisions about biomass collection or aggregation? *(Unsang impormasyon ang inyong kinahanglan mahibal-an una bago magdesisyon sa pagkolekta o pagtigom sa biomass?).*	
10	How can GIS mapping help in monitoring biomass sources, cooperative hubs, and collection points? *(Sa inyong tan-aw, unsaon pagtabang sa GIS mapping sa pag-monitor kung asa gikan ang biomass, asa ang cooperative hubs, ug asa ang collection points?).*	
11	What biomass residues are already used by farmers, rice millers, or local households? *(Unsang mga biomass residue ang gigamit na sa farmers, rice millers, o mga balay sa inyong lugar?).*	
12	How do you know if a biomass residue is still practically available for collection or selling? *(Giunsa ninyo pagkahibalo kung ang biomass residue available pa gyud para kolektahon o ibaligya?).*	
13	What storage, moisture, or quality problems affect biomass use? *(Unsang mga problema sa storage, pagkabasa, o quality ang makaapekto sa paggamit sa biomass?).*	

14	Who should check or validate biomass records before they are listed or reported? *(Kinsa ang angay mo-check o mo-validate sa biomass records bago kini ibutang sa listahan o i-report?).*	
15	What biomass reports would be useful for LGU planning, cooperative monitoring, or buyer coordination? *(Unsang klase sa biomass reports ang makatabang sa LGU planning, pag-monitor sa cooperative, o pag-coordinate sa buyers?).*	

APPENDIX D
SYSTEM DESIGN AND DEVELOPMENT DOCUMENTATION MATRIX

Design and development research is appropriate when a study builds and evaluates a product, model, process, or system (Richey & Klein, 2007). In this study, the system output is supported by the Biomass Availability Information Framework and the cooperative-based validation workflow. *(Ang design and development research angay gamiton kung ang study naghimo ug nag-evaluate og product, model, process, o system (Richey & Klein, 2007). Sa kini nga study, ang paghimo sa system gibase sa Biomass Availability Information Framework ug sa proseso sa pag-check ug pag-validate sa biomass data pinaagi sa cooperative.).*

Objective Component	**Corresponding Module / Feature**	**Development Output**
Standardized digital encoding and management of biomass data. *(Parehas ug klaro nga paagi sa pag-encode ug pag-manage sa biomass data).*	Biomass Data Collection and Management Module	Forms and database records for biomass type, estimated volume, source location, source contributor, availability period, and quality observations. *(Mga form ug database records para sa klase sa biomass, estimated amount, asa kini gikan, kinsa ang naghatag sa information, kanus-a kini available, ug notes bahin sa quality.).*
Practical biomass availability assessment. *(Pag-check kung available pa ba gyud ang biomass para kolektahon o gamiton).*	Biomass Availability Assessment Module	Records for existing use, available quantity, availability status, collection window, distance concern, storage condition, moisture observation, and practical availability notes. *(Mga record kung gigamit na ba ang*

		biomass, pila pa ang available, unsa ang current status, kanus-a kini pwede kolektahon, kung layo ba, unsa ang storage condition, kung basa ba, ug uban pang notes kung makuha ug magamit pa ba gyud kini.).
Cooperative-based data entry, validation, and aggregation workflows. *(Pag-encode, pag-check, ug pagtigom sa biomass data pinaagi sa cooperative).*	Cooperative Validation Workflow Module	Role-based workflow for cooperative hub managers, rice mill personnel, NGO coordinators, LGU representatives, and authorized users to encode, check, approve, reject, or update biomass records. *(Proseso nga nagtakda kung kinsa sa cooperative hub managers, rice mill personnel, NGO coordinators, LGU representatives, ug authorized users ang pwede mo-encode, mo-check, mo-approve, mo-reject, o mo-update sa biomass records.).*
GIS-based mapping, visualization, and monitoring of biomass sources. *(Paggamit sa GIS map para makita ug ma-monitor kung asa gikan ang biomass).*	GIS-Based Biomass Mapping and Visualization Module	Map view showing biomass-producing locations, cooperative hubs, collection points, rice mills, and relevant facilities. *(Map nga nagpakita kung asa makita ang biomass sources, cooperative hubs, collection points, rice mills, ug uban pang related nga facilities.).*

Centralized biomass inventory and quality information management. *(Usa ka system para ma-manage ang biomass inventory ug quality information).*	Biomass Inventory and Quality Management Module	Inventory records for stock level, storage capacity, availability status, quality notes, aggregation updates, and record changes. *(Inventory records nga nagpakita sa amount sa stock, storage capacity, availability status, quality notes, updates sa pagtigom sa biomass, ug mga changes sa records.).*
Structured reporting, transaction tracking, and information exchange. *(Organisadong reporting, pag-monitor sa transactions, ug pag-share sa information).*	Reports, Transaction Monitoring, Audit Logging, and Marketplace Coordination Module	Dashboards, reports, biomass listings, buyer inquiries, transaction records, inventory updates, and user activity logs. *(Mga dashboard, reports, biomass listings, buyer inquiries, transaction records, inventory updates, ug records sa activities sa users.).*
LGU and institutional data use. *(Paggamit sa biomass data sa LGU ug uban pang institutions).*	LGU Reporting and Monitoring Output	Summarized reports that may help update biomass resource information, identify source areas, and support waste-to-value planning in Butuan City. *(Summary reports nga makatabang sa pag-update sa biomass resource information, pag-ila kung asa gikan ang biomass, ug pagsuporta sa waste-to-value planning sa Butuan City.).*

APPENDIX E
SYSTEM EVALUATION QUESTIONNAIRE

The evaluation is limited to Functional Suitability and Usability because these are the selected ISO/IEC 25010 software quality characteristics for AgriCycle Butuan. Functional Suitability checks whether the system provides the needed functions, while Usability checks whether intended users can understand and use the system (International Organization for Standardization, 2011). *(Ang evaluation naka-focus lang sa Functional Suitability ug Usability kay mao kini ang gipiling ISO/IEC 25010 software quality characteristics para sa AgriCycle Butuan. Ang Functional Suitability tan-awon kung naa ba sa system ang mga functions nga kinahanglan, samtang ang Usability tan-awon kung sayon ba sabton ug gamiton ang system sa intended users (International Organization for Standardization, 2011).*

Rating	Interpretation
5	Strongly Agree *(Uyon kaayo)*
4	Agree *(Uyon)*
3	Neutral
2	Disagree *(Dili uyon)*
1	Strongly Disagree *(Dili gyud uyon)*

A. Functional Suitability

This section checks whether the system supports biomass availability information management, cooperative validation, GIS mapping, aggregation monitoring, and marketplace coordination. *(Kini nga section tan-awon kung ang system adunay mga functions nga makatabang sa pag-manage sa biomass availability information, pag-check ug pag-validate sa data pinaagi sa cooperative, GIS mapping, pag-monitor sa pagtigom sa biomass, ug pag-coordinate sa marketplace.).*

No.	Statement	5	4	3	2	1
1	The system provides the necessary functions for recording biomass availability.					

	(Ang system adunay mga functions nga kinahanglan para ma-record kung unsang biomass ang available.).					
2	The system allows authorized users to encode biomass type, estimated volume, source location, availability period, and quality observations. *(Ang system nagtugot sa authorized users nga mo-encode sa klase sa biomass, estimated amount, asa kini gikan, kanus-a kini available, ug information bahin sa quality.).*					
3	The system supports standardized digital encoding of biomass data. *(Ang system adunay parehas ug klaro nga paagi sa digital nga pag-encode sa biomass data.).*					
4	The system records whether biomass is available, already used, reserved, unavailable, or pending validation. *(Ang system maka-record kung ang biomass available pa, gigamit na, gitagana na para sa lain, dili available, o wala pa na-check ug na-validate.).*					
5	The system allows authorized users to record existing uses of biomass residues. (*Ang system nagtugot sa authorized users nga i-record kung giunsa na paggamit ang biomass residues.).*					
6	The system allows users to record seasonality or availability periods. *(Ang system nagtugot sa users nga i-record kung unsang panahon o kanus-a available ang biomass.).*					
7	The system allows users to record storage conditions, moisture conditions, and quality observations. *(Ang system nagtugot sa users nga i-record ang storage condition, kung basa ba ang biomass, ug information bahin sa quality.).*					
8	The system allows authorized users to update and validate biomass records.					

	(Ang system nagtugot sa authorized users nga mo-update ug mo-check o mo-validate sa biomass records.).					
9	The system supports cooperative-based data entry and aggregation workflows. *(Ang system makatabang sa pag-encode sa data ug pagtigom sa biomass pinaagi sa cooperative.).*					
10	The system allows users to monitor biomass inventory status. *(Ang system nagtugot sa users nga ma-monitor ang status sa biomass inventory.).*					
11	The system provides GIS-based mapping of biomass source locations. *(Ang system adunay GIS map nga nagpakita kung asa gikan ang biomass.).*					
12	The system displays cooperative hubs, collection points, rice mills, or biomass-generating areas on the map. *(Ang system nagpakita sa map kung asa ang cooperative hubs, collection points, rice mills, ug mga lugar nga adunay biomass.).*					
13	The system allows users to publish or view biomass marketplace listings. *(Ang system nagtugot sa users nga mag-post o motan-aw sa biomass listings sa marketplace.).*					
14	The system allows institutional buyers or authorized users to browse available biomass resources. *(Ang system nagtugot sa institutional buyers o authorized users nga motan-aw kung unsang biomass resources ang available.).*					
15	The system allows inquiry submission or expression of interest for available biomass resources. *(Ang system nagtugot sa users nga mag-send og inquiry o magpakita nga interesado sila sa available nga biomass.).*					

16	The system records inquiry status, inventory updates, aggregation records, and user activities. *(Ang system maka-record sa status sa inquiries, inventory updates, records sa pagtigom sa biomass, ug activities sa users.).*					
17	The system provides dashboard summaries or reports for biomass availability, inventory, aggregation, marketplace inquiries, and transactions. *(Ang system adunay dashboard summaries o reports bahin sa available nga biomass, inventory, pagtigom sa biomass, marketplace inquiries, ug transactions.).*					
18	The system provides information useful for LGU monitoring and waste-to-value planning. *(Ang system makahatag og information nga makatabang sa LGU monitoring ug waste-to-value planning.).*					
19	The system helps distinguish theoretical biomass estimates from practical biomass availability. *(Ang system makatabang sa pagpakita sa kalainan sa estimated biomass ug sa biomass nga naa ug pwede pa gyud kolektahon o gamiton.).*					
20	Overall, the system functions match the intended purpose of AgriCycle Butuan. *(Sa kinatibuk-an, ang mga functions sa system angay sa tumong sa AgriCycle Butuan.).*					

B. Usability

This section checks whether intended users can understand, learn, navigate, and use the system. This is important because the system will be used by people with different levels of digital familiarity. *(Kini nga section tan-awon kung sayon ba sabton, tun-an, pangitaon ang mga features, ug gamiton ang system. Importante kini kay lain-lain ang experience ug kahibalo sa users sa paggamit og digital systems.).*

No.	Statement	5	4	3	2	1

21	The system is easy to understand. *(Sayon ra sabton ang system.)*					
22	The system is easy to learn for its intended users. *(Sayon ra tun-an kung unsaon paggamit sa system.)*					
23	The labels, buttons, menus, and forms are clear. *(Klaro ug sayon sabton ang labels, buttons, menus, ug forms.)*					
24	The dashboard is organized and easy to navigate. *(Organisado ang dashboard ug sayon ra pangitaon ang kinahanglan nga features.)*					
25	The biomass data entry form is easy to complete. *(Sayon ra sulatan ug kumpletuhon ang biomass data entry form.)*					
26	The availability status options are easy to understand. *(Sayon ra sabton ang mga options nga nagpakita kung available ba ang biomass o dili.)*					
27	The validation workflow is easy to follow. *(Sayon ra sundon ang proseso sa pag-check ug pag-validate sa data.)*					
28	The inventory and aggregation pages are easy to use. *(Sayon ra gamiton ang inventory ug aggregation pages.)*					
29	The GIS map is easy to interpret. *(Sayon ra sabton ang impormasyon nga makita sa GIS map.)*					
30	The marketplace listings are easy to browse and understand. *(Sayon ra tan-awon ug sabton ang biomass listings sa marketplace.)*					
31	The inquiry or transaction status is easy to follow. *(Sayon ra ma-check kung unsa na ang status sa inquiry o transaction.)*					

32	The reports and dashboard summaries are presented clearly. *(Klaro ug sayon sabton ang reports ug summaries sa dashboard.)*					
33	The system allows users to complete tasks with minimal confusion. *(Sayon ra mahuman sa users ang ilang buhaton sa system nga dili kaayo malibog.)*					
34	Overall, the system provides a user-friendly experience for biomass information management and marketplace coordination. *(Sa overall, sayon ug komportable gamiton ang system para sa pag-manage sa biomass information ug pag-coordinate sa marketplace.)*					

APPENDIX F

OPEN-ENDED SYSTEM FEEDBACK FORM

This form supports interpretation of the system evaluation results by allowing respondents to explain useful, unclear, difficult, or missing parts of the system. Thematic grouping may be used for the responses (Braun & Clarke, 2006). *(Kini nga form maghatag og chance sa respondents nga ma-share kung unsang parts sa system ang useful, dili klaro, lisod gamiton, o kulang pa. Ang ilang mga tubag mahimong ipundok base sa parehas nga tema (Braun & Clarke, 2006).*

No.	Question	Response
1	What part of AgriCycle is most useful for biomass data collection and information management? *(Unsang part sa AgriCycle ang pinaka gamit para sa pagkolekta ug pag-manage sa biomass information?)*	
2	What part of the system is most useful for checking practical biomass availability? *(Unsang part sa system ang pinaka gamit para ma-check kung available pa ba gyud ang biomass para kolektahon o gamiton?)*	
3	What part of the system is most useful for cooperative-based validation? *(Unsang part sa system ang pinaka gamit para sa pag-check ug pag-validate sa biomass data pinaagi sa cooperative?)*	
4	What part of the system is most useful for aggregation or inventory monitoring? *(Unsang part sa system ang pinaka gamit para sa pagtigom sa biomass o pag-monitor sa inventory?)*	
5	What part of the GIS mapping feature is useful? *(Unsang part sa GIS mapping feature ang*	

	nakatabang para sa inyo?)	
6	What part of the GIS mapping feature is unclear or difficult to use? *(Unsang part sa GIS mapping feature ang dili klaro o lisod gamiton?)*	
7	What information should be added to biomass records, inventory records, or marketplace listings? *(Unsang impormasyon ang kinahanglan pa idugang sa biomass records, inventory records, o marketplace listings?)*	
8	What part of the system is confusing, difficult, or unnecessary? *(Unsang part sa system ang makalibog, lisod gamiton, o dili kaayo kinahanglan?)*	
9	What features should be improved before finalizing the system? *(Unsang feature ang kinahanglan pa i-improve bago mahuman ang final system?)*	
10	What recommendation can improve AgriCycle for stakeholders in Butuan City? *(Unsay inyong ma-suggest para mas mapalambo ug mas makatabang ang AgriCycle sa mga involved nga tawo o grupo sa Butuan City?)*	

APPENDIX G
DATA ANALYSIS AND THEMATIC GROUPING TEMPLATE

This template supports descriptive interpretation of Likert-type responses and thematic grouping of open-ended answers. Mean scores will describe the level of agreement, while open-ended responses will be grouped according to repeated ideas and concerns (Braun & Clarke, 2006; Likert, 1932). *(Kini nga template gamiton sa pag-analyze sa mga tubag sa Likert scale ug sa pag-group sa open-ended answers. Ang mean score magpakita kung unsa ka dako ang pag-uyon sa respondents, samtang ang open-ended answers i-group base sa mga ideas o concerns nga balik-balik makita sa ilang mga tubag (Braun & Clarke, 2006; Likert, 1932).*

Mean Range	**Interpretation**
4.21-5.00	Very High / Strongly Agree
3.41-4.20	High / Agree
2.61-3.40	Moderate / Neutral
1.81-2.60	Low / Disagree
1.00-1.80	Very Low / Strongly Disagree

Theme	**Related Comments / Evidence**	**Frequency**	**Interpretation / Action Taken**
Biomass data collection issues. *(Mga problema sa pagkolekta sa biomass data).*			
Record standardization. *(Pagparehas sa paagi sa pag-record).*			
Information			

fragmentation. *(Nagkatag nga biomass information).*			
Reporting delays. *(Pagdugay sa pag-report).*			
Existing biomass uses. *(Kasamtangang paggamit sa biomass).*			
Practical biomass availability. *(Kung available pa ba gyud ang biomass).*			
Seasonality and collection timing. *(Panahon kung kanus-a available ug makolekta ang biomass).*			
Transportation and location concerns. *(Mga problema sa transport ug lokasyon).*			
Storage, moisture, and quality concerns. *(Mga problema sa storage, pagkabasa, ug quality).*			
Data validation concerns. *(Mga problema sa pag-check ug*			

pag-validate sa data).			
GIS mapping concerns. *(Mga problema o concerns sa GIS mapping).*			
Inventory monitoring needs. *(Mga kinahanglan para sa pag-monitor sa inventory).*			
Marketplace listing needs. *(Mga kinahanglan sa marketplace listings).*			
LGU and BEDP data use. *(Paggamit sa data para sa LGU ug BEDP).*			
Usability and interface clarity. *(Kasayon sa paggamit ug pagkaklaro sa system interface).*			
Suggested improvements. *(Mga gi-suggest nga improvements).*			

APPENDIX H

BIOMASS DATA FIELD CHECKLIST

(Checklist sa mga Information nga Kinahanglan i-record para sa Biomass)

This checklist supports the system design by listing the data fields needed to apply the Biomass Availability Information Framework. The fields are limited to information management and coordination. They do not require laboratory testing, route optimization, power plant sizing, or financial computation. *(Kini nga checklist naglista sa mga information nga kinahanglan i-record para magamit ang Biomass Availability Information Framework. Ang mga information naka-focus lang sa pag-manage ug pag-coordinate sa biomass data. Dili kinahanglan ani ang laboratory testing, pagplano sa pinakamaayong ruta, power plant sizing, o financial computation.).*

Data Field Group	**Information to Record**
Biomass Identification. *(Pag-ila sa Biomass).*	Biomass type; crop source; residue form; source contributor; barangay or collection point. *(Klase sa biomass; unsang tanom kini gikan; porma sa residue; kinsa ang naghatag sa information; barangay o collection point.).*
Quantity and Unit. *(Kadaghanon ug Unit).*	Estimated quantity; unit of measurement; basis of estimate; date recorded. *(Estimated amount; unit sama sa kilo o tonelada; unsay basehan sa estimate; ug petsa nga gi-record.).*
Location. *(Lokasyon).*	Barangay; farm, mill, or hub location; coordinates if available; nearest aggregation point. *(Barangay; lokasyon sa farm, mill, o hub; coordinates kung naa; ug pinakaduol nga lugar nga pwede tiguman ang biomass.).*
Availability	Availability period; collection window; current status; expected next update. *(Kanus-a available ang biomass; kanus-a kini pwede kolektahon; current status; ug kanus-a sunod i-update ang record.).*
Existing Use. *(Kasamtangang Paggamit).*	Used internally; used for drying or heating; used for cooking; used for charcoal; composted; left in field; burned; unused; other local use. *(Gigamit sa tag-iya; gigamit sa pagpauga o pagpainit; gigamit sa pagluto; gigamit para sa charcoal;*

	gihimong compost; gibiyaan sa uma; gisunog; wala gigamit; o uban pang paggamit sa lugar.).
Practical Availability Status. *(Kung Available Pa ba Gyud ang Biomass).*	Available; pending validation; reserved; already used; unavailable; unsuitable; for follow-up. *(Available; wala pa na-check; gitagana na para sa lain; gigamit na; dili na makuha; dili angay gamiton; o kinahanglan pa i-check balik.).*
Storage and Quality	Storage condition; moisture observation; contamination notes; visible condition; handling concern. *(Storage condition; kung basa ba ang biomass; kung naay sagol nga hugaw o lain nga material; makita nga condition; ug problema sa pagdala o pag-handle.).*
Transport and Access	Distance concern; road access; hauling availability; collection difficulty notes. *(Problema sa kalayo; kahimtang sa dalan; kung naay available nga panghakot; ug mga problema sa pagkolekta.).*
Validation	Encoded by; checked by; validation date; validation result; remarks. *(Kinsa ang nag-encode; kinsa ang nag-check; petsa sa pag-check; resulta sa validation; ug remarks o notes.).*
Marketplace and Coordination	Listing status; buyer inquiry; inquiry status; transaction note; stakeholder contact channel. *(Status sa listing; inquiry sa buyer; status sa inquiry; note bahin sa transaction; ug paagi sa pagkontak sa involved nga tawo o grupo.).*
LGU and Reporting	Report category; barangay summary; biomass source map; update log; audit trail. *(Klase sa report; summary sa barangay; mapa kung asa gikan ang biomass; record sa mga updates; ug history sa mga changes sa system.).*

APPENDIX I

GANTT CHART

Agricycle: A Community–Based Biomass Information Management and Market Linkage System

Project Duration: 11 weeks

Legend	Phase Summary		Planned Bar	

WBS	Phase	Task / Activity	Key Deliverable / Output	Assigned Member
1	**Preparation Phase**	**Phase Summary**	**Product backlog, approved requirements list, data dictionary, user roles, and development plan**	**All Members**
1.1	Preparation Phase	Review problem background and local biomass information needs	Problem background notes and initial scope confirmation	All Members
1.2	Preparation Phase	Organize user needs and define core data fields	User needs list and biomass data field list	All Members
1.3	Preparation Phase	Identify user roles and access levels	Role list for encoder, validator, LGU, hub, buyer, and administrator users	All Members
1.4	Preparation Phase	Prepare process flow, data dictionary, and initial interface plan	Process flow, data dictionary, interface plan, and initial backlog	All Members
M1	**Milestone**	**Backlog and requirements approved**	**Approved requirements list and sprint–ready backlog**	**All Members**
2	**Sprint 1**	**Core System Foundations Summary**	**Working user access, biomass source records, and biomass data records**	**All Members**
2.1	Sprint 1	Sprint planning, module design, and task breakdown	Sprint 1 plan and task board	All Members
2.2	Sprint 1	Develop login and role–based access	Login module and controlled user access	All Members
2.3	Sprint 1	Develop biomass source profiles	Source profile records for biomass contributors, mills, hubs, and related sources	All Members
2.4	Sprint 1	Develop biomass data encoding, standard fields, required field controls, and basic record management	Working biomass data records with required fields and basic management functions	All Members
2.5	Sprint 1	Internal check, feedback recording, and revision	Sprint 1 review notes and revised core data functions	All Members
M2	**Milestone**	**Core record functions completed**	**User access, biomass source records, and biomass data records completed**	**All Members**
3	**Sprint 2**	**Validation and GIS Visibility Summary**	**Working validation workflow, data quality controls, and GIS visualization functions**	**All Members**
3.1	Sprint 2	Sprint planning, module design, and task breakdown	Sprint 2 plan and task board	All Members
3.2	Sprint 2	Develop validation status, validation notes, and approval flow	Validation workflow with status updates and review notes	All Members
3.3	Sprint 2	Develop data quality checks, status rules, required fields, and	Data quality controls for completeness, consistency, and credibility	All Members
3.3	Sprint 2	Develop data quality checks, status rules, required fields, and consistent units	Data quality controls for completeness, consistency, and credibility	All Members
3.4	Sprint 2	Develop map interface for source locations, rice mills, cooperative hubs, and collection points	GIS map with filterable biomass–related locations	All Members
3.5	Sprint 2	Internal check, feedback recording, and revision	Sprint 2 review notes and revised validation/GIS functions	All Members
M3	**Milestone**	**Validation and GIS functions completed**	**Validated records, data quality controls, and GIS visibility completed**	**All Members**
4	**Sprint 3**	**Aggregation and Inquiry Coordination Summary**	**Working aggregation, inventory, listing, and inquiry functions**	**All Members**
4.1	Sprint 3	Sprint planning, module design, and task breakdown	Sprint 3 plan and task board	All Members
4.2	Sprint 3	Develop hub inventory, aggregation records, storage status, and quality notes	Hub inventory and aggregation monitoring functions	All Members
4.3	Sprint 3	Develop validated listings	Listing function limited to validated available biomass records	All Members
4.4	Sprint 3	Develop buyer inquiry form and inquiry status tracking	Buyer inquiry records and inquiry status monitoring	All Members
4.5	Sprint 3	Internal check, feedback recording, and revision	Sprint 3 review notes and revised aggregation/inquiry functions	All Members
M4	**Milestone**	**Aggregation and inquiry functions completed**	**Inventory, listing, and inquiry coordination completed**	**All Members**
5	**Sprint 4**	**Reports, Dashboard, Logs, and Final Integration Summary**	**Integrated system ready for internal testing and user evaluation**	**All Members**
5.1	Sprint 4	Sprint planning, module design, and task breakdown	Sprint 4 plan and task board	All Members
5.2	Sprint 4	Develop reports, dashboard summaries, and exportable records where applicable	Reports, dashboard summaries, and applicable exportable records	All Members
5.3	Sprint 4	Develop transaction logs, activity logs, and audit logs	Traceable system activity and audit logs	All Members
5.4	Sprint 4	Conduct interface review and full system checking	Interface review notes and full system checking results	All Members
5.5	Sprint 4	Record feedback and apply final sprint revisions	Revision log and integrated system ready for internal testing and user evaluation	All Members
M5	**Milestone**	**Integrated system completed**	**AgriCycle system ready for internal testing and user evaluation**	**All Members**

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